

**AN INTRODUCTION TO PARTIAL DIFFERENTIAL EQUATIONS
WITH AN EMPHASIS ON HARMONIC FUNCTIONS**

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ABSTRACT. In these notes we introduce the concept of harmonic functions. Our ultimate goal is to study solutions to the Dirichlet problem in domains $\Omega \subset \mathbb{R}^n$. To realize this goal, we shall prove fundamental results applying to harmonic functions, which include the mean value formula and Harnack's inequality. We attempt to use methods of proof which yield analogous results for subharmonic and superharmonic functions. We also place a special emphasis on the regularity properties of harmonic functions. To tackle the existence of solutions, we will work in Sobolev spaces whilst making heavy use of techniques from functional analysis. In particular, we assume familiarity with spectral theory and Hilbert spaces.

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Part 1. Theory and Lectures

1. INTRODUCTION AND DEFINITIONS

Throughout this text, we shall only work in $n \geq 2$ dimensions. A *domain* is defined to be an open and connected subset Ω of $\mathbb{R}^n$. An open set Ω is said to be *pre-compact* if it is bounded. Thus, an open set Ω is pre-compact if and only if $\overline{\Omega}$ is compact. Given any set $V \subseteq \mathbb{R}^n$, we will agree to write $\Omega \Subset V$ if Ω is bounded and $\overline{\Omega} \subseteq V$. Here, $\overline{\Omega}$ denotes the topological closure of Ω in $\mathbb{R}^n$. Unless stated otherwise, any set is assumed to be non-empty.

Definition 1. Let $\Omega \subseteq \mathbb{R}^n$ be open and let $u \in C^2(\Omega)$. We say that u is harmonic at a point $x_0 \in \Omega$ if

$$\Delta u(x_0) := \sum_{j=1}^n \frac{\partial^2 u}{\partial x_j^2}(x_0) = 0.$$

If u is harmonic at all points $x_0 \in \Omega$, we say that u is harmonic in Ω . The differential operator Δ is called the Laplace operator.

It is not difficult to see that Δ is defined for all functions $u \in C^2(\Omega)$. In fact, if $u \in C^2(\Omega)$, then Δu returns a continuous function on Ω . Thus, we can choose to view Δ as a linear operator

$$\Delta : C^2(\Omega) \rightarrow C(\Omega), \quad u \mapsto \Delta u.$$

The set of all harmonic functions in Ω shall henceforth be denoted by $\mathfrak{H}(\Omega)$. Since we have only defined Δ for functions $u \in C^2(\Omega)$, it is clear that $\mathfrak{H}(\Omega) \subseteq C^2(\Omega)$. Notice also that $\mathfrak{H}(\Omega)$ is never empty since constant functions are harmonic.

Remark 1.1. In $n = 1$ the Laplacian reduces to the operator $\Delta = \frac{d^2}{dx^2}$. Then, the only solutions to $\Delta u = 0$ are the linear equations $y = ax + b$. This certainly shows that the case $n = 1$ is far from interesting. For this reason, we restrict ourselves to the case $n \geq 2$.

Although we have currently only defined *harmonicity* for functions in $C^2(\Omega)$, we will later give an equivalent definition that will allow us to *extend* our definition to functions that are merely continuous on Ω . For the moment, however, we will have to content ourselves with C^2 -functions.

1.1. Subharmonic and Super-harmonic functions. We continue to assume that Ω is a non-empty open subset of $\mathbb{R}^n$. A continuous function $u : \Omega \rightarrow \mathbb{R}$ is said to be *subharmonic* in Ω if

$$u(x) \leq \oint_{\partial B(x,r)} u(y) \, d\sigma(y), \quad \text{for all balls } B(x,r) \Subset \Omega. \quad (1.1)$$

Here, $d\sigma$ denotes the $(n - 1)$ -dimensional surface measure on the sphere $\partial B(x, r)$. Given a general measure space $(X, \mathfrak{M}, \mu)$, a set $E \in \mathfrak{M}$ of finite positive measure, and a function $f \in L^1(E)$, we define

$$\oint_E f \, d\mu := \frac{1}{\mu(E)} \int_E f \, d\mu.$$

Intuitively, $\oint_{\partial B(x, r)} f \, d\sigma$ represents the average of f over the sphere $\partial B(x, r)$. We will denote the space of all subharmonic functions in Ω by $\mathfrak{Sub}(\Omega)$. Note that, unlike with $\mathfrak{H}(\Omega)$, subharmonic functions are *not* assumed to be differentiable. Similarly, we say that a function $u \in C(\Omega)$ is *super-harmonic* in Ω if

$$\oint_{\partial B(x, r)} u(y) \, d\sigma(y) \leq u(x), \quad \text{for all balls } B(x, r) \Subset \Omega. \quad (1.2)$$

Note that u is super-harmonic in Ω if and only if $-u$ is subharmonic. Naturally, we will denote by $\mathfrak{Sup}(\Omega)$ the set of super-harmonic functions $\Omega \rightarrow \mathbb{R}$.

1.2. Motivation from Complex Analysis. To the reader familiar with complex function theory, it will shortly become apparent that harmonic functions have much of the properties that one finds in holomorphic functions. This is no coincidence and we will demonstrate exactly why this phenomenon occurs. Informally, we will show that holomorphic functions *are* harmonic. Thus, properties of holomorphic functions are really properties of *harmonic*.

Let now $\Omega \subseteq \mathbb{C}$ be a non-empty open set and suppose that $f : \Omega \rightarrow \mathbb{C}$ is holomorphic on Ω . Given any point $z \in \Omega$, we know that the limit

$$\lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{C}}} \frac{f(z + h) - f(z)}{h} = f'(z)$$

exists. In particular, the limit should be equal to h if we take the limit along any path with $h \rightarrow 0$. Let $u(x, y) = \Re f$ and $v(x, y) = \Im f$; for $z = x + iy$ we calculate

$$\begin{aligned} f'(z) &= \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{C}}} \frac{f(z + h) - f(z)}{h} = \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{f(z + h) - f(z)}{h} \\ &= \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{u(x + h, y) + iv(x + h, y) - u(x, y) - iv(x, y)}{h} \\ &= \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{u(x + h, y) - u(x, y)}{h} + i \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{v(x + h, y) - v(x, y)}{h} \\ &= \frac{\partial u}{\partial x}(z) + i \frac{\partial v}{\partial x}(z). \end{aligned}$$

On the other hand, a similar argument gives

$$\begin{aligned}
f'(z) &= \lim_{\substack{h \rightarrow 0 \\ \Re h = 0}} \frac{f(z+h) - f(z)}{h} \\
&= \lim_{h \rightarrow 0} \frac{u(x, y+h) + iv(x, y+h) - u(x, y) - iv(x, y)}{ih} \\
&= -i \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{u(x, y+h) - u(x, y)}{h} + \lim_{\substack{h \rightarrow 0 \\ h \in \mathbb{R}}} \frac{v(x, y+h) - v(x, y)}{h} \\
&= \frac{\partial v}{\partial y}(z) - i \frac{\partial u}{\partial y}(z).
\end{aligned}$$

Combining these two expressions yields

$$\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \equiv \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} \quad \text{in } \Omega.$$

Equating the real and imaginary parts of this expression then gives

$$\begin{cases} \frac{\partial u}{\partial x} \equiv \frac{\partial v}{\partial y} & \text{in } \Omega, \\ \frac{\partial u}{\partial y} \equiv -\frac{\partial v}{\partial x} & \text{in } \Omega. \end{cases} \quad (1.3)$$

These are called the *Cauchy-Riemann* equations and must be satisfied by any holomorphic function $f : \Omega \rightarrow \mathbb{C}$. Recall that holomorphisms are smooth; we thus know that all the partial derivatives of both exist.

Proposition 1.1. *Let $\Omega \subseteq \mathbb{C}$ be non-empty and open. If $f : \Omega \rightarrow \mathbb{C}$ is holomorphic in Ω , then $\Re f$ and $\Im f$ are harmonic functions on Ω , when Ω is viewed as a subset of $\mathbb{R}^2$.*

Proof. We only prove the claim for $u = \Re f$ and leave $\Im f$ as an exercise. Invoking the Cauchy-Riemann equations in (1.3) immediately gives the desired result:

$$\begin{aligned}
\Delta u &\equiv \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \equiv \frac{\partial}{\partial x} \left(\frac{\partial v}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial x} \right) \\
&\equiv \frac{\partial^2 v}{\partial x \partial y} - \frac{\partial^2 v}{\partial y \partial x} \\
&\equiv 0.
\end{aligned}$$

□

Thus, the real and imaginary parts of holomorphic functions are harmonic. It then makes sense to view f as “complex harmonic”, whenever f is holomorphic in an open set Ω .

2. THE MEAN VALUE FORMULAS

We have given a definition for harmonic functions in terms of the differential operator Δ . Our goal in this section is to give a useful characterization of harmonic functions in terms of integrals. More precisely, we would like to relate our definition of harmonic functions to the identities in (1.1)-(1.2). Again, Ω continues to denote a non-empty open subset of $\mathbb{R}^n$, with $n \geq 2$.

Theorem 2.1. *Let $u \in C^2(\Omega)$ be given; the following statements are equivalent.*

- (1) *u is subharmonic in Ω , i.e. (1.1) holds true.*
- (2) *$-\Delta u \leq 0$ in Ω .*

Proof. First assume that u is subharmonic in Ω according to the definition in (1.1). Suppose for a contradiction that $-\Delta u$ is positive at some point $x_0 \in \Omega$. Equivalently, $\Delta u(x_0) < 0$. Choose now an open ball $B(x_0, r) \Subset \Omega$ such that $\Delta u < 0$ on the closure of the ball. Define a function $\eta(\rho)$ on $(0, r)$ according to the formula

$$\eta(\rho) := \oint_{\partial B(x, \rho)} u(y) \, d\sigma(y). \quad (2.1)$$

Since $u \in C^2(\Omega)$, we are free to compute

$$\eta'(\rho) = \frac{\partial}{\partial \rho} \left[\frac{1}{n\omega_n \rho^{n-1}} \int_{\partial B(x, \rho)} u(y) \, d\sigma(y) \right] \quad (2.2)$$

$$= \frac{\partial}{\partial \rho} \left[\frac{1}{n\omega_n} \int_{\partial B(0, 1)} u(x_0 + \rho z) \, d\sigma(z) \right] \quad (2.3)$$

$$= \frac{1}{n\omega_n} \int_{\partial B(0, 1)} \nabla u(x_0 + \rho z) \cdot z \, d\sigma(z) \quad (2.4)$$

$$= \frac{1}{n\omega_n \rho^{n-1}} \int_{\partial B(x_0, \rho)} \nabla u(y) \cdot \frac{y - x_0}{\rho} \, d\sigma(y) \quad (2.5)$$

$$= \frac{1}{n\omega_n \rho^{n-1}} \int_{\partial B(x_0, \rho)} \nabla u(y) \cdot \nu(y) \, d\sigma(y) \quad (2.6)$$

$$= \frac{1}{n\omega_n \rho^{n-1}} \int_{B(x_0, \rho)} \Delta u(y) \, dy. \quad (2.7)$$

Here, ω_n denotes the volume of the unit n -ball and $\nu(y)$ is the outwards pointing unit normal on $\partial B(x_0, r)$. Since $\Delta u < 0$ on the closure of $B(x_0, r)$, the above tells us that $\eta'(\rho)$ is *strictly negative* for $0 < \rho < r$. Hence, η is strictly decreasing for $0 < \rho < r$. By continuity of u :

$$u(x_0) = \lim_{\rho \searrow 0} \oint_{\partial B(x_0, \rho)} u(y) \, d\sigma(y) = \lim_{\rho \searrow 0} \eta(\rho).$$

Since η is strictly decreasing for $0 < \rho < r$, we subsequently have

$$\oint_{\partial B(x, \rho)} u(y) d\sigma(y) = \eta(\rho) < u(x_0),$$

which contradicts (1.1). We conclude that $\Delta u \geq 0$ on Ω .

Conversely, suppose that $\Delta u \geq 0$ on Ω . Given a point $x_0 \in \Omega$, let r be such that $B(x_0, r) \Subset \Omega$. Define η as above; our calculations show that η is increasing for $0 < \rho < r$. Again, by the continuity of u , we have

$$u(x_0) = \lim_{\rho \searrow 0} \oint_{\partial B(x_0, \rho)} u(y) d\sigma(y) = \lim_{\rho \searrow 0} \eta(\rho).$$

Since η is increasing for $0 < \rho < r$, we must have

$$u(x_0) \leq \oint_{\partial B(x_0, \rho)} u(y) d\sigma(y)$$

for all $0 < \rho < r$. Taking a limit via the dominated convergence theorem, we conclude that $u(x_0) \leq \oint_{\partial B(x_0, r)} u(y) d\sigma(y)$. Hence, (1.1) is satisfied in Ω . $\square$

Remark 2.1. It would have been natural to define a function to be subharmonic whenever $-\Delta u \leq 0$. However, this would have at least required u to be twice differentiable on its domain. By our theorem above, we now see that the definition we gave in (1.1) is more general than the condition “ $-\Delta u \leq 0$ ”. Indeed, we allow functions that are merely continuous to be subharmonic.

In light of Theorem 2.1, we have an analogous characterization of super-harmonic functions.

Corollary 2.2. *Let $u \in C^2(\Omega)$ be given; the following statements are equivalent.*

- (1) *u is super-harmonic in Ω , i.e. (1.2) holds true.*
- (2) *$-\Delta u \geq 0$ in Ω .*

Proof. Apply the previous theorem to $v = -u$. $\square$

2.1. The Mean Value Property for Harmonic Functions. Now we show how our characterization of subharmonic and super-harmonic functions applies to functions that are harmonic. First, suppose that $u \in C^2(\Omega)$. By Theorem 2.1 and Corollary 2.2, we see that u is harmonic if and only if $u \in \mathfrak{Sub}(\Omega) \cap \mathfrak{Sup}(\Omega)$.¹ This observation immediately implies the following.

Theorem 2.3. *Let Ω be a non-empty open subset of $\mathbb{R}^n$ and suppose $u \in C^2(\Omega)$. The following statements are equivalent:*

- (1) *$u \in \mathfrak{H}(\Omega)$, i.e. u is harmonic in Ω ;*

¹Note that the requirement $u \in C^2(\Omega)$ is absolutely crucial here.

(2) for all open balls $B(x, r) \Subset \Omega$ there holds

$$u(x) = \oint_{\partial B(x, r)} u(y) \, d\sigma(y). \quad (2.8)$$

If either (1) or (2) holds true, then for all balls $B(x, r) \Subset \Omega$:

$$u(x) = \oint_{B(x, r)} u(y) \, dy. \quad (2.9)$$

Equations (2.8)-(2.9) are called the mean value formulas for u .

Proof. The only statement that does not follow at once from Theorem 2.1 and Corollary 2.2 is (2.9). To verify this, let $B(x_0, r) \Subset \Omega$ and calculate

$$\begin{aligned} \oint_{B(x, r)} u(y) \, dy &= \frac{1}{\omega_n r^n} \int_0^r \left(\int_{\partial B(x, \rho)} u(y) \, d\sigma(y) \right) d\rho \\ &\stackrel{(2.8)}{=} \frac{1}{r^n} \int_0^r n \rho^{n-1} u(x) \, d\rho \\ &= u(x). \end{aligned}$$

This proves the theorem. $\square$

In light of this theorem, it makes sense for us to *extend* the definition of a harmonic function to all functions in $C(\Omega)$ via equation (2.8). Let us formally make this generalization below.

Definition 2. Let Ω be a non-empty open subset of $\mathbb{R}^n$. A function $u \in C(\Omega)$ is said to be (*generalized*) *harmonic* in Ω if it satisfies (2.8) for all open balls $B(x, r) \Subset \Omega$. Note that by the proof we have just given, u will also satisfy (2.9) in Ω .

We will later show that all harmonic² functions in $C(\Omega)$ are necessarily of class $C^2(\Omega)$, and therefore harmonic in the classical sense. In fact, these functions will always turn out to be smooth! It is also possible to prove that any function $u \in L^1(\Omega)$ for which (2.8) is satisfied must also be smooth, but we will not give this result.

3. THE DIRICHLET PROBLEM AND POISSON'S FORMULA

Let $\Omega \subseteq \mathbb{R}^n$ be a bounded domain. Given a continuous function $g \in C(\partial\Omega)$, the Dirichlet problem for g on Ω is the boundary value problem

$$\begin{cases} \Delta u \equiv 0 & \text{in } \Omega, \\ u \equiv g & \text{on } \partial\Omega. \end{cases} \quad (3.1)$$

²That is, functions that are harmonic according to our generalized definition.

Given another function $f : \Omega \rightarrow \mathbb{R}$, the Poisson problem for f and g involves finding a twice differentiable function u on Ω such that

$$\begin{cases} \Delta u \equiv f & \text{in } \Omega, \\ u \equiv g & \text{on } \partial\Omega. \end{cases} \quad (3.2)$$

In both cases, we are typically looking for solutions $u \in C^2(\Omega) \cap C(\overline{\Omega})$. For the most part, we will handle only the Dirichlet problem (3.1). In later sections, we will actually show that a $C^2(\Omega) \cap C(\overline{\Omega})$ solution to either (3.1) or (3.2) must be unique, whenever it exists. Our first step in this direction is to show that the Dirichlet problem (3.1) has a solution whenever Ω is an open ball.

3.1. The Poisson Formula. Our goal is to solve the Dirichlet problem (3.1) when Ω is an open ball of the form $B_r = B(x_0, r)$. After a translation, it is clearly enough to consider the case $x_0 = 0$. Since the result we present in this subsection is rather tedious to prove, it will be given without a rigorous justification. For this reason, we choose to define the following function without motivation:

$$\Pi : B_r \times \partial B_r \rightarrow \mathbb{R}, \quad (x, y) \mapsto \frac{r^2 - |x|^2}{n\omega_n r |x - y|^n}. \quad (3.3)$$

The function $\Pi(x, y)$ is called the *Poisson kernel* for the ball B_r . Note that, for fixed $y \in \partial B_r$, the function $\Pi(\cdot, y)$ is smooth in its first variable (since x can only take values away from y). Moreover, $\Pi(x, y)$ behaves similarly to an “approximation to the identity” near a point y on the boundary of ∂B_r . With this in mind, the following theorem (first proven by Schwarz in 1872) is not too hard to accept.

Theorem 3.1. *Let $r > 0$ and $g \in C(\partial B_r)$ be given. Define a function $u : B_r \rightarrow \mathbb{R}$ according to the formula:*

$$u(x) := \int_{\partial B_r} g(y) \Pi(x, y) \, d\sigma(y). \quad (3.4)$$

Then, $u \in C^\infty(B_r)$ is harmonic in B_r (i.e. $\Delta u \equiv 0$ in B_r). Moreover, u extends to a continuous function $\overline{\Omega} \rightarrow \mathbb{R}$ after setting $u := g$ on ∂B_r .

Thus, we have solved the Dirichlet problem (3.1) for balls. Although we will not prove the theorem above, we frequently invoke it by way of giving nice proofs of future fundamental results.

4. THE MAXIMUM PRINCIPLES

We have already considered harmonic functions and their relation to the mean value property (2.8). In fact, we proved that a function $u \in C^2(\Omega)$ is harmonic in Ω if and only if (2.8) (and hence (2.9)) is satisfied there. A priori, it is natural to

attempt to weaken the condition in (2.9) to obtain an even more general definition of harmonic functions. After all, (2.9) is assumed to hold for *all* balls $B(x, r) \Subset \Omega$. With this in mind, we state the *local mean value property*.

Definition 3. Let Ω be a non-empty open set and $u \in C(\Omega)$. We say that u satisfies the *local mean value property* in Ω if, for every $x \in \Omega$, there exists $r^* > 0$ such that

$$u(x) = \oint_{\partial B(x, r)} u(y) \, d\sigma(y) \quad (4.1)$$

for all $0 < r < r^*$.

Remark 4.1. Suppose that u satisfies the local mean value property in Ω . Given a point $x \in \Omega$, we can find $r^* > 0$ so small that

$$u(x) = \oint_{\partial B(x, r)} u(y) \, d\sigma(y)$$

for all $0 < r < r^*$. For any such value of r , (4.1) yields

$$\oint_{B(x, r)} u(y) \, dy = \frac{1}{\omega_n r^n} \int_0^r \left(\int_{\partial B(x, \rho)} u(y) \, d\sigma(y) \right) d\rho \quad (4.2)$$

$$= \frac{u(x)}{r^n} \int_0^r n\rho^{n-1} \, d\rho \quad (4.3)$$

$$= u(x). \quad (4.4)$$

The above is frequently called the *local mean value property for balls*.

Clearly, generalized harmonic functions satisfy the local mean value property. Indeed, these functions satisfy the mean value property in Ω by definition. What may be of surprise is that the local mean value property is equivalent to the mean value property. In fact, every definition of harmonicity we have given is identical. However, we do not yet have the tools to prove this.

Theorem 4.1 (The Strong Maximum Principle). *Let Ω be a bounded domain³ and suppose that $u \in C(\overline{\Omega})$ satisfies one of the following:*

- (1) $u \in C^2(\Omega)$ and $\Delta u \equiv 0$ in Ω ;
- (2) u satisfies the mean value property (2.8);
- (3) u satisfies the local mean value property (4.1).

If there exists a point $x_0 \in \Omega$ such that

$$u(x_0) = \max_{\overline{\Omega}} u,$$

then u is constant on $\overline{\Omega}$.

³A non-empty, open and connected subset of $\mathbb{R}^n$.

Proof. By earlier results and observations, assumptions (1) and (2) both imply (3). Thus, it is enough to assume (3).

Let m denote the maximum of u on $\overline{\Omega}$ and assume x_0 is an interior point of Ω such that $u(x_0) = m$. Consider now the subset

$$\mathcal{X} := \{x \in \Omega : u(x) = m\}.$$

Clearly, $\mathcal{X}$ is non-empty. By the continuity of u , it is closed in Ω . If we can show that $\mathcal{X}$ is open in Ω , then by connectedness it will follow that $\mathcal{X} = \Omega$. Of course, this would force u to be constant in Ω , and hence on $\overline{\Omega}$. To this end, let $x \in \Omega$ be given and choose $r^* > 0$ so small that (4.1) holds for all $0 < r < r^*$. Without loss of generality, assume that $B(x, r^*) \Subset \Omega$. For any $0 < r < r^*$, we apply Remark 4.1 to obtain

$$m = u(x) = \int_{B(x,r)} u(y) \, dy.$$

Hence, $0 = \int_{B(x,r)} (u(x) - u(y)) \, dy$ where $u(x) \geq u(y)$ in Ω . Thus, $u(y) \equiv m$ for almost all $y \in B(x, r)$. The continuity of u then forces u to be constant (equal to m) on $B(x, r)$ whence $B(x, r) \subseteq \mathcal{X}$. Therefore, $\mathcal{X}$ is clopen and equal to Ω . $\square$

If Ω is not necessarily connected, we can break it down into its connected components. Applying the strong maximum principle to each of these components, we obtain the so-called *weak maximum principle*.

Corollary 4.2. *Let Ω be a non-empty and bounded open set in $\mathbb{R}^n$. If $u \in C(\overline{\Omega})$ satisfies one of the following criteria*

- (1) $u \in C^2(\Omega)$ and $\Delta u \equiv 0$ in Ω ,
- (2) u satisfies the mean value property (2.8),
- (3) u satisfies the local mean value property (4.1);

then

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u.$$

Observe that every criterion listed in the strong maximum principle and the corollary is unaffected if we replace u with $-u$. Therefore, we obtain an *analogous strong minimum principle* for functions u satisfying one of (1)-(3).

Theorem 4.3. *Let Ω be a bounded open set and suppose that $u \in C(\overline{\Omega})$ satisfies one of the following:*

- (1) $u \in C^2(\Omega)$ and $\Delta u \equiv 0$ in Ω ;
- (2) u satisfies the mean value property (2.8);
- (3) u satisfies the local mean value property (4.1).

Then

$$\max_{\partial\Omega} u = \min_{\partial\Omega} u.$$

If Ω is connected and there exists a point $x_0 \in \Omega$ such that

$$u(x_0) = \max_{\overline{\Omega}} u,$$

then u is constant on $\overline{\Omega}$.

4.1. Maximum Principles for Subharmonic Functions. The strong maximum (resp. minimum) principle continues to hold if u is merely assumed to be subharmonic (resp. super-harmonic) in the domain Ω . In this subsection, we justify this claim.

Theorem 4.4. *Let Ω be a bounded domain in $\mathbb{R}^n$ and suppose $u \in C(\overline{\Omega})$ is subharmonic in Ω . If there exists an interior point $x_0 \in \Omega$ such that $u(x_0) = \max_{\overline{\Omega}} u$, then $u \equiv m$ in $\overline{\Omega}$.*

Proof. We once again consider the non-empty closed subset of Ω :

$$\mathcal{X} := \{x \in \Omega : u(x) = m\}.$$

Let $x \in \Omega$ be given and choose $r > 0$ so small that $B(x, r) \Subset \Omega$. Because u is subharmonic in Ω ,

$$m = u(x) \leq \int_{\partial B(x, r)} u(y) \, d\sigma(y).$$

In fact, an elementary calculation similar to that in Remark 4.1 shows that

$$m = u(x) \leq \int_{B(x, r)} u(y) \, dy.$$

Hence, $0 \leq \int_{B(x, r)} (u(y) - u(x)) \, dy \leq 0$ where $u(x) \geq u(y)$ in Ω . Thus, $u(y) \equiv m$ for almost all $y \in B(x, r)$. The continuity of u then forces u to be constant (equal to m) on $B(x, r)$ whence $B(x, r) \subseteq \mathcal{X}$. Therefore, $\mathcal{X}$ is clopen and equal to Ω . $\square$

As before, we deduce an easy corollary.

Corollary 4.5. *Let Ω be a non-empty and bounded open set in $\mathbb{R}^n$. Suppose that $u \in C(\overline{\Omega})$ is subharmonic in Ω , then*

$$\max_{\overline{\Omega}} u = \max_{\partial\Omega} u.$$

Recalling that a function $u \in C(\Omega)$ is super-harmonic if and only if $-u$ is subharmonic, we readily deduce the minimum principles below.

Theorem 4.6. *Let Ω be a non-empty bounded open set in $\mathbb{R}^n$ and assume $u \in C(\overline{\Omega})$ is subharmonic in Ω . Then,*

$$\min_{\overline{\Omega}} u = \min_{\partial\Omega} u.$$

Suppose in addition that Ω is connected. If there exists a point $x_0 \in \Omega$ such that $u(x_0) = \min_{\overline{\Omega}} u$, then $u \equiv u(x_0)$ on $\overline{\Omega}$.

4.2. Uniqueness to the Poisson Problem. As promised, we show that there exists at most a single $C^2(\Omega) \cap C(\overline{\Omega})$ solution to the Poisson problem (3.2). Of course, this implies uniqueness for the Dirichlet problem (3.1). Do note that our proof heavily depends upon the max/min principles from the previous subsection.

Theorem 4.7. *Let Ω be a non-empty open bounded subset of $\mathbb{R}^n$. Given functions $f \in C(\Omega)$ and $g \in C(\partial\Omega)$, there exists at most a single solution $u \in C^2(\Omega) \cap C(\overline{\Omega})$ to the Poisson problem (3.2). In particular, the Dirichlet problem (3.1) has at most one solution.*

Proof. Let u and v be solutions to the Poisson equation (3.2). Then, $w := u - v$ is a harmonic function in Ω that vanishes on $\partial\Omega$. Since w is both subharmonic and super-harmonic, we can apply the maximum and minimum principles to obtain

$$0 = \max_{\partial\Omega} w = \max_{\overline{\Omega}} w$$

and

$$0 = \min_{\partial\Omega} w = \min_{\overline{\Omega}} w.$$

This completes the proof. □

5. REGULARITY AND DERIVATIVE ESTIMATES

In the previous section we showed that, for a function $u \in C^2(\Omega)$, the mean value and harmonic properties are equivalent. Afterwards, we realized that one could generalize the notion of harmonicity by declaring a function $u \in C(\Omega)$ to be harmonic if it satisfies the mean value property (i.e. (2.8)) in Ω . Extrapolating further, we introduced the even weaker local mean value property. Our next result establishes the equivalence of these three concepts.

Theorem 5.1 (Regularity Theorem). *Let Ω be a non-empty open subset of $\mathbb{R}^n$ and assume $u \in C(\Omega)$ satisfies the local mean value property in Ω . Then we necessarily have $u \in C^\infty(\Omega) \cap \mathfrak{H}(\Omega)$. More precisely, u is harmonic and smooth in Ω .*

Proof. Fix a point $x_0 \in \Omega$ and choose $r^* > 0$ such that

$$u(x) = \oint_{\partial B(x_0, r^*)} u(y) \, d\sigma(y)$$

for all $0 < r < r^*$. Take also r^* so small that $B(x_0, r^*) \Subset \Omega$. It is enough to show that u is smooth and harmonic in this open ball. After a translation, we may assume without loss of generality that $x_0 = 0$. Now fix $0 < r < r^*$; clearly, u is continuous on $\overline{B(0, r)}$. Moreover, u satisfies the local mean value property in $B(0, r)$.

Let now $v \in C^\infty(B(0, r)) \cap C(\overline{B(0, r)})$ be the solution to the Dirichlet problem

$$\begin{cases} \Delta v \equiv 0 & \text{in } B(0, r), \\ v \equiv u & \text{on } \partial B(0, r) \end{cases}$$

obtained by convolving u with the Poisson kernel Π . Since v is harmonic in $B(0, r)$, it satisfies the (local) mean value property in $B(0, r)$. In particular, $u - v$ satisfies the local mean value property in $B(0, r)$. By Corollary 4.2, we get $u - v \leq 0$. On the other hand, we can apply the same argument to get $v - u \leq 0$. It follows that $u \equiv v$. That is, u is locally equal to a smooth harmonic function. The claim follows. $\square$

The theorem above immediately implies the following equivalence result.

Corollary 5.2. *Let $u \in C(\Omega)$ be given. The following statements are equivalent:*

- (1) $u \in C^2(\Omega)$ and $\Delta u \equiv 0$ in Ω ;
- (2) u satisfies the mean value property in Ω ;
- (3) u satisfies the local mean value property in Ω .

If any one of the above hold, then $u \in C^\infty(\Omega)$.

Thus, we see that harmonic functions are always smooth in their domains. In particular, all partial derivatives exist.

5.1. L^1 -Derivative Estimates and Applications. Let Ω be a non-empty open subset of $\mathbb{R}^n$ and let u be harmonic in Ω (according to any definition). By virtue of the last corollary, we might as well assume that u is a smooth solution to $\Delta u \equiv 0$ in Ω . Subsequently, it is natural to ask whether being harmonic gives us any control on the derivatives of u in Ω . The answer here is a *partial* yes. We have control, but only locally.

For now, let $x \in \Omega$ be given and suppose that $r > 0$ is such that $B(x, r) \Subset \Omega$. In particular, $B(x, r/2) \Subset \Omega$. Let $j = 1, \dots, n$ be an index and note that, by commutativity of the partial derivatives, the function u_{x_i} is also harmonic in Ω . The mean value formula then gives

$$\begin{aligned} u_{x_j}(x) &= \frac{2^n}{\omega_n r^n} \int_{B(x, r/2)} u_{x_i}(y) \, dy \\ &= \frac{2^n}{\omega_n r^n} \int_{\partial B(x, r/2)} u(y) \nu^i \, d\sigma(y). \end{aligned}$$

Therefore,

$$|u_{x_j}(x)| \leq \frac{2^n}{\omega_n r^n} \int_{\partial B(x, r/2)} |u(y)| d\sigma(y) \quad (5.1)$$

$$\leq \frac{2n}{r} \|u\|_{L^\infty(\partial B(x, r/2))}. \quad (5.2)$$

Now, choose $y \in \partial B(x, r/2)$. It is easy to see that $B(y, r/2) \subseteq B(x, r)$. Invoking once more the mean value property, we discover that

$$|u(y)| \leq \frac{2^n}{r^n \omega_n} \int_{B(y, r/2)} |u(y)| dy \leq \frac{2^n}{r^n \omega_n} \|u\|_{L^1(B(x, r))}.$$

Thus, $\|u\|_{L^\infty(\partial B(x, r/2))} \leq \frac{2^n}{r^n \omega_n} \|u\|_{L^1(B(x, r))}$. Hence,

$$|u_{x_j}(x)| \leq \frac{2n}{r} \cdot \frac{2^n}{r^n \omega_n} \|u\|_{L^1(B(x, r))} = \frac{C}{r^{n+1}} \|u\|_{L^1(B(x, r))},$$

for a constant $C > 0$ independent of u and $r > 0$. Arguing by induction on the order of the derivatives, we obtain the following:

Theorem 5.3 (Derivative Estimates). *Let u be harmonic in Ω (according to any of our definitions). For any multiindex α , and any ball $B(x, r) \Subset \Omega$, there holds*

$$|\partial^\alpha u(x)| \leq \frac{C}{r^{n+|\alpha|}} \|u\|_{L^1(B(x, r))}. \quad (5.3)$$

Here, $C > 0$ depends only on n and $|\alpha|$.

For the sake of clarity and elegance, we omit the details of the inductive step.

5.1.1. Entire Harmonic Functions. A function $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is called *entire harmonic* if $u \in C^2(\mathbb{R}^n)$ and $\Delta u \equiv 0$ in $\mathbb{R}^n$. That is, if u is harmonic in all of $\mathbb{R}^n$. It is a relatively elementary result in complex analysis that every bounded entire holomorphic function $\mathbb{C} \rightarrow \mathbb{C}$ is constant. This statement continues to hold for entire harmonic functions. This reinforces the fact that complex analytic results are really applications of harmonic function theory.

Theorem 5.4. *Any bounded entire harmonic function is constant.*

Proof. Let $u \in C^2(\mathbb{R}^n)$ be bounded and harmonic in all of space. Let α be any multiindex of degree 1. Given any point $x \in \mathbb{R}^n$ and $r > 0$, (5.3) guarantees the existence of a positive constant $C > 0$ independent of r such that

$$|\partial^\alpha u(x)| \leq \frac{C}{r^{n+1}} \|u\|_{L^1(B(x, r))}.$$

Let $M > 0$ be an upper bound for u in $\mathbb{R}^n$; after possibly correcting the constant $C > 0$ we get

$$|\partial^\alpha u(x)| \leq \frac{C}{r^{n+1}} \cdot Mr^n = \frac{C}{r}.$$

Letting $r \rightarrow \infty$, we see that ∇u vanishes in all of space. Thus, u is constant. $\square$

5.1.2. The Fundamental Theorem of Algebra. Let $P : \mathbb{C} \rightarrow \mathbb{C}$ be a non-constant polynomial, i.e. suppose $\deg P \geq 1$. We claim that $P(z)$ must have a root in $\mathbb{C}$. Otherwise $P(z) \neq 0$ for all $z \in \mathbb{C}$ whence $P(z)^{-1}$ is entire holomorphic. Since $P(z)$ is continuous and

$$\lim_{|z| \rightarrow \infty} \frac{1}{P(z)} = 0,$$

it is easily seen that $Q(z) := P(z)^{-1}$ is bounded in $\mathbb{C}$. In particular, both $\Re Q$ and $\Im Q$ are bounded harmonic functions in $\mathbb{R}^2$. It follows that both $\Re Q$ and $\Im Q$ are constant in $\mathbb{R}^2$. Consequently, $P(z)$ is constant in $\mathbb{C}$, which is a contradiction.

6. HARNACK'S THEOREMS

Harnack's theorems are cornerstones of harmonic function theory. His inequality gives us useful local weak monotonicity, whilst his other theorems deal with the local convergence of harmonic functions. We now prove the former.

Theorem 6.1 (Harnack's Inequality). *Let Ω be an open and bounded subset of $\mathbb{R}^n$. For every open and connected set $V \Subset \Omega$, there exists a constant $C > 0$ such that*

$$\sup_V u \leq C \inf_V u$$

for all non-negative $u \in \mathfrak{H}(\Omega)$.

Proof. Since $\bar{V}$ is compact, we can define

$$r := \frac{\text{dist}(\bar{V}, \partial\Omega)}{4} > 0.$$

Then, for any $x \in \bar{V}$, the open ball $B(x, 2r)$ is completely contained in Ω . Using that V is connected, we cover V by finitely many balls $B_1, \dots, B_N$, each of radius $r/2$, such that $B_i \cap B_{i-1}$ is non-empty for every index $i > 1$. Suppose now that $x, y \in V$ satisfy $|x - y| \leq r$. Then, $B(y, r) \subseteq B(x, 2r)$. It follows that

$$u(y) = \int_{B(y,r)} u(z) \, dz = \frac{1}{\omega_n r^n} \int_{B(y,r)} u(z) \, dz.$$

Since u is non-negative, this yields

$$\begin{aligned} u(y) &\leq \frac{1}{\omega_n r^n} \int_{B(x, 2r)} u(z) \, dz = 2^n \oint_{B(x, 2r)} u(z) \, dz \\ &= 2^n u(x). \end{aligned}$$

Iteratively applying this argument, we see that $u(y) \leq 2^{N+1}u(x)$ for all $x, y \in V$. This inequality yields the desired inequality. $\square$

We now move onto the so called Harnack convergence theorems, the first of which is highly reminiscent of a result from complex analysis.

Theorem 6.2. *Let Ω be a non-empty open subset of $\mathbb{R}^n$ and $\{u_k\}_{k \in \mathbb{N}}$ a sequence of harmonic functions in Ω . If u_k converges locally uniformly to a function u on Ω , then u is harmonic in Ω . Moreover, $\partial^\alpha u_k \rightarrow \partial^\alpha u$ locally uniformly on Ω for all multiindices α .*

Proof. Clearly, $u \in C(\Omega)$. To establish that u harmonic, it is enough to show that u satisfies the mean value property in Ω . Let $B(x, r) \Subset \Omega$ be given; since each u_k is harmonic in Ω we have

$$u_k(x) = \oint_{\partial B(x, r)} u_k(y) \, d\sigma(y).$$

Letting $k \rightarrow \infty$ and applying the uniform convergence theorem yields

$$\begin{aligned} u(x) &= \lim_{k \rightarrow \infty} u_k(x) = \lim_{k \rightarrow \infty} \oint_{\partial B(x, r)} u_k(y) \, d\sigma(y) \\ &= \oint_{\partial B(x, r)} u(y) \, d\sigma(y). \end{aligned}$$

Hence, $u \in \mathfrak{H}(\Omega)$. For the second part, let α be a multiindex and fix a compact set $K \subset \Omega$. Cover K by finitely many open balls, each compactly contained in Ω . By hypothesis, u_k will converge uniformly to u on each of these balls. Let $\delta > 0$ be a Lebesgue number for K and this cover by open balls. Applying the derivative estimates with $x \in K$ and $B(x, \delta)$, the claim follows. $\square$

Theorem 6.3 (Harnack's Principle). *Let $\Omega \subseteq \mathbb{R}^n$ be a domain and assume $\{u_k\}_{k \in \mathbb{N}}$ is a sequence of harmonic functions in Ω with:*

$$u_1 \leq u_2 \leq \cdots \leq u_k \leq \cdots$$

Then, exactly one of the following statements hold.

- (1) $u_k(x) \nearrow \infty$ as $k \rightarrow \infty$, for each $x \in \Omega$.
- (2) u_k converges locally uniformly in Ω . In particular, u_k converges pointwise to a harmonic function in Ω .

Remark 6.1. Harnack's principle can be interpreted as a monotone convergence theorem for harmonic functions.

Proof of Harnack's Principle. Assume that there exists at least one point $y \in \Omega$ such that $\sup_{k \in \mathbb{N}} u_k(y) < \infty$. In particular, $(u_k(y))_{k \in \mathbb{N}}$ converges in $\mathbb{R}$. Fix a compact subset K of Ω and assume, without loss of generality, that $y \in K$.⁴ Let $V \Subset \Omega$ be a connected open set containing the compact space K .⁵ Fix $\ell \in \mathbb{N}$ and observe that

$$u_{k+\ell} - u_k \geq 0 \text{ in } \Omega, \quad \forall k \geq 1.$$

Let Λ be an open and bounded subset of Ω such that $V \Subset \Lambda$.⁶ An application of Harnack's inequality gives, for some constant $C > 0$,

$$(u_{k+\ell}(x) - u_k(x)) \leq C (u_{k+\ell}(y) - u_k(y)) \xrightarrow{k, \ell \rightarrow \infty} 0.$$

The above holds for all $x \in V$, and hence all $x \in K$. Thus, u_k converges uniformly on $V \supset K$. $\square$

6.1. A Compactness Result for Harmonic Families. Let's take a moment to recall that Arzela-Ascoli theorem from real analysis. Let $\Omega \subseteq \mathbb{R}^n$ be a domain and let $\{f_k\}_{k \in \mathbb{N}}$ be a sequence that is both

- locally equibounded on Ω ;
- locally equicontinuous on Ω .

Then, there is a subsequence $\{f_{n_k}\}$ of f that converges *locally uniformly* on Ω . Harnessing this theorem, we will be able to establish an analogous compactness criterion for families of harmonic functions. We make this rigorous below.

Theorem 6.4. *Let $\Omega \subseteq \mathbb{R}^n$ be a domain and assume that $\{f_k\}_{k \in \mathbb{N}}$ is family in $C^2(\Omega)$ such that:*

- (1) *the f_{n_k} are all harmonic in Ω ;*
- (2) *the sequence is locally equibounded.*

Then, there is a subsequence $\{f_{n_k}\}_{k \in \mathbb{N}}$ that converges locally uniformly to a harmonic function on Ω .

Proof. In view of the Arzela-Ascoli and Harnack theorems, it is enough to check that the sequence $\{f_k\}_{k \in \mathbb{N}}$ is locally equicontinuous on Ω . By locality, it is enough to

⁴Otherwise, we simply replace K by $K \sqcup \{y\}$ which is also compact.

⁵To see that such a set V actually exists, we choose finitely many open balls $B_1, \dots, B_N$, each compactly contained in Ω , whose union covers K . Since Ω is connected, it is path connected. Therefore, we can choose these B_i such that $B_i \cap B_{i-1} \neq \emptyset$ for all indices $i = 2, \dots, N$. Then, the union of these balls has compact closure in Ω , is connected, and contains K .

⁶Take Λ to be the intersection of Ω with some large open ball.

prove that the sequence is equicontinuous on a compact ball $\bar{B} \subset \Omega$. Thus, let us consider two balls

$$B := B_\rho(y) \quad \text{and} \quad B' := B_{\rho+r}(y)$$

with $r > 0$ and $B' \Subset \Omega$. Now, given $x \in B$ and any harmonic function $u \in C^2(\Omega)$, we have $\overline{B(x, r)} \subseteq B'$. In particular,

$$\begin{aligned} |\partial_i u(x)| &= \frac{1}{\omega_n r^n} \left| \int_{B(x, r)} u_{x_i} \, dm \right| = \frac{1}{\omega_n r^n} \left| \int_{\partial B(x, r)} u \nu^i \, dS \right| \\ &\leq \frac{1}{\omega_n r^n} \int_{\partial B(x, r)} |u| \, dS \\ &\leq \frac{n}{r} \sup_{\bar{B}'} |u|. \end{aligned}$$

Especially, for each index k ,

$$|\partial_i u_k(x)| \leq \frac{n}{r} \sup_{\bar{B}'} |u_k|, \quad \forall x \in B.$$

Thus, we have the following gradient bound:

$$\sup_{x \in B} |\nabla u_k(x)| \leq \frac{C}{r} \sup_{\bar{B}'} |u|.$$

By the mean value theorem, we see that the $\{u_k\}_k$ are equicontinuous on B , and therefore on any small compact ball centered at y . Since $y \in \Omega$ was arbitrary, this establishes the local equicontinuity. $\square$

7. PERRON'S METHOD

We have already solved the Dirichlet problem (3.1) when Ω is an open ball in $\mathbb{R}^n$. The next step is to extend this existence result to more general domains $\Omega \subset \mathbb{R}^n$. To achieve this, we will follow a method due to Oskar Perron. For the remainder of this section, we make the following standing assumption.

Assumption. We denote by Ω an open and bounded domain in $\mathbb{R}^n$, for $n \geq 2$. We also let $g : \partial\Omega \rightarrow \mathbb{R}$ be any bounded function.

Before introducing Perron's method, we would to make some basic observations regarding subharmonic functions. The proofs are relatively easy and roughly illustrated below.

- (1) If $u \in \mathbf{Sub}(\Omega)$ and there exists $x_0 \in \Omega$ such that $u(x_0) = \sup_\Omega u$, then u is constant in Ω .
- (2) Let $u \in \mathbf{Sub}(\Omega)$ be continuous on $\bar{\Omega}$ and suppose $v \in C(\bar{\Omega})$ is harmonic in Ω . If $u \leq v$ on $\partial\Omega$, then $u \leq v$ on Ω .

- (3) If $u_1, u_2 \in \mathfrak{Sub}(\Omega)$ then so is $\max(u_1, u_2)$.
- (4) Let $u \in \mathfrak{Sub}(\Omega)$ and $\bar{u} \in C(\Omega)$. If $\Delta \bar{u} \equiv 0$ in some domain $U \Subset \Omega$, and is equal to u in $\Omega \setminus U$, then $\bar{u} \in \mathfrak{Sub}(\Omega)$.

Let us now verify these facts. For (1), we see that the standard proof of the maximum principle applies. For (2), we show that $w := u - v$ is subharmonic in Ω . If this were true, then the maximum principle would force

$$\max_{\bar{\Omega}}(u - v) \leq \max_{\partial\Omega}(u - v) \leq 0$$

whence the result follows. Let $B(x, r) \Subset \Omega$ be given and so that

$$u(x) \leq \int_{\partial B(x, r)} u(y) d\sigma(y) \quad \text{and} \quad v(x) = \int_{\partial B(x, r)} v(y) d\sigma(y).$$

Combining these yields

$$\begin{aligned} w(x) = u(x) - v(x) &\leq \int_{\partial B(x, r)} (u(y) - v(y)) d\sigma(y) \\ &= \int_{\partial B(x, r)} w(y) d\sigma(y). \end{aligned}$$

Thus, $w \in \mathfrak{Sub}(\Omega)$ as was asserted. For (3), we use a similar argument. Let $B(x, r)$ be compactly contained in Ω and let $v := \max(u_1, u_2)$. For $j = 1$ or $j = 2$, we calculate

$$u_j(x) \leq \int_{\partial B(x, r)} u_j(y) d\sigma(y) \leq \int_{\partial B(x, r)} v(y) d\sigma(y).$$

Thus, $v(x) \leq \int_{\partial B(x, r)} v(y) d\sigma(y)$ whence $\max(u_1, u_2) \in \mathfrak{Sub}(\Omega)$. We will omit the justification of (4).

7.1. Describing Perron's Method. We continue to assume that Ω is a bounded domain in $\mathbb{R}^n$, for $n \geq 2$. Again, g is a given *bounded* function $\partial\Omega \rightarrow \mathbb{R}$. We define

$$\mathcal{S}_g := \{v \in \mathfrak{Sub}(\Omega) \cap C(\bar{\Omega}) : v|_{\partial\Omega} \leq g\}. \quad (7.1)$$

Note that $\mathcal{S}_g$ is non-empty as it contains any constant map c satisfying $c \leq g$ on $\partial\Omega$. Moreover, given $v \in \mathcal{S}_g$, the maximum principle for subharmonic functions asserts that

$$\max_{\bar{\Omega}} v \leq \max_{\partial\Omega} v \leq \max_{\partial\Omega} g < \infty.$$

Thus, the elements of $\mathcal{S}_g$ are uniformly bounded on $\bar{\Omega}$. In the hopes of solving the Dirichlet problem (3.1) for this Ω and g , we define the *Perron solution* as follows:

$$u(x) := \sup_{v \in \mathcal{S}_g} v(x). \quad (7.2)$$

By our earlier observations, this gives us a well defined map $u : \Omega \rightarrow \mathbb{R}$. For the sake of a sanity check, we want to show that this gives us the *unique* solution to the Dirichlet problem on Ω when we know that one exists. Suppose for the moment that w is a $C^2(\Omega) \cap C(\bar{\Omega})$ solution to the Dirichlet problem

$$\begin{cases} \Delta w \equiv 0 & \text{in } \Omega, \\ w \equiv g & \text{on } \partial\Omega. \end{cases}$$

In particular, we may assume that $g \in C(\partial\Omega)$. We now ask ourselves whether the Perron solution (7.2) gives us this function w . First, note that $w \in \mathcal{S}_g$ and thus u must satisfy $u(x) \geq w(x)$ in Ω . On the other hand, if $v \in \mathcal{S}_g$, then observation (2) tells us that $v \leq w$ in Ω . Hence,

$$u(x) = \sup_{v \in \mathcal{S}_g} v(x) \leq w(x)$$

for all $x \in \Omega$.

Theorem 7.1 (Perron). *Under the given conditions, the function u prescribed by (7.2) is in $C^2(\Omega)$ and satisfies $\Delta u \equiv 0$ in Ω .*

We will omit the proof of this result, as the argument is convoluted and difficult to write out. We will, however, sketch out the proof. Choose an open ball $B(x, r) \Subset \Omega$ and, for this x , let $\{u_k\}_{k \in \mathbb{N}}$ be a sequence in $\mathcal{S}_g$ converging to $u(x)$ in $\mathbb{R}$ as $k \rightarrow \infty$. Without loss of generality, assume that $u_k(x) \nearrow u(x)$. By observation (3), we can replace each u_k with $\max(u_1, \dots, u_k)$ whilst preserving the pointwise convergence at x and the subharmonicity in Ω . For each $k \in \mathbb{N}$, let $\bar{u}_k$ be the unique solution to the problem

$$\begin{cases} \Delta \bar{u}_k \equiv 0 & \text{in } B(x, r), \\ \bar{u}_k \equiv u_k & \text{on } \partial B(x, r). \end{cases}$$

We can then extend $\bar{u}_k$ to a continuous function on Ω that agrees with u_k in $\Omega \setminus B(x, r)$. Again invoking the observation in (2), we have $u_k \leq \bar{u}_k$ on Ω . Clearly, $\bar{u}_k \in \mathcal{S}_g$ for every $k \in \mathbb{N}$. Thus,

$$u_k(x) \leq \bar{u}_k(x) \leq u(x).$$

Letting $k \rightarrow \infty$, it follows that $\bar{u}_k(x) \rightarrow u(x)$. By construction, the sequence $\{\bar{u}_k\}_{k \in \mathbb{N}}$ is monotone increasing on $\Omega \setminus B(x, r)$. Because every $w_k := \bar{u}_{k+1} - \bar{u}_k$ is a smooth solution to

$$\begin{cases} \Delta w_k \equiv 0 & \text{in } B(x, r), \\ w_k \geq 0 & \text{on } \partial B(x, r); \end{cases}$$

it follows from the maximum principle for harmonic functions that $\bar{u}_{k+1} - \bar{u}_k \geq 0$ on $B(x, r)$, and thus on all of Ω . Hence, $\{\bar{u}_k\}_{k \in \mathbb{N}}$ is increasing on Ω . An application

of Harnack's principle shows that there exists a harmonic function $\bar{u}$ in $B(x, r)$ such that $\bar{u}_k \rightarrow \bar{u}$ locally uniformly on $B(x, r)$. Especially,

$$\bar{u}(x) = \lim_{k \rightarrow \infty} u_k(x) = u(x).$$

It is now enough to show that $\bar{u} \equiv u$ on this ball. To this end, we fix a point $y \in B(x, r)$. Repeating the argument above, and making the suitable corrections, we can choose a monotone increasing sequence of functions $\{v_k\}_{k \in \mathbb{N}}$ in $\mathcal{S}_g$, each harmonic in $B(x, r)$, such that $\bar{u}_k \leq v_k$ and $v_k(y) \rightarrow u(y)$. Again, there must exist a harmonic function v in $B(x, r)$ such that $v_k \rightarrow v$ locally uniformly on $B(x, r)$. In particular, $v(y) = u(y)$. Since $\bar{u}_k \leq v_k$ for all k , we have $\bar{u} \leq v$ on all of $B(x, r)$. However, $v_k \leq u$ certainly implies $v \leq u$ in $B(x, r)$. Thus,

$$u(x) = \bar{u}(x) \leq v(x) \leq u(x)$$

whence $v(x) = u(x)$. Since $\bar{u} - v \leq 0$ in $B(x, r)$ and $\bar{u}(x) = v(x)$, the strong maximum principle gives $\bar{u} \equiv v$ in $B(x, r)$. Especially, we get that $\bar{u}(y) = u(y)$.

7.2. Boundary Conditions. The Perron construction ensures the existence of a function $u : \Omega \rightarrow \mathbb{R}$ that is harmonic in Ω , but it is not immediately clear that this u will satisfy the boundary condition $u \equiv g$. To rectify this issue, we introduce the notion of a barrier, which is due to Lebesgue.

Definition 4. A function $\varphi \in C(\bar{\Omega}) \cap \mathfrak{Sub}(\Omega)$ is said to be a *barrier* at a point $z \in \partial\Omega$

$$\varphi(z) = 0 \quad \text{and} \quad \varphi(z) < 0 \text{ on } \bar{\Omega} \setminus \{z\}.$$

A point $z \in \partial\Omega$ is called *regular* if there exists a barrier φ at z .

Our important result is the following.

Theorem 7.2. *Suppose that g is continuous at a point $z \in \partial\Omega$. If z is a regular point, then*

$$\lim_{\Omega \ni x \rightarrow z} u(x) = g(z),$$

where u is the Perron solution defined by (7.2). In particular, we have solved the Dirichlet problem when $g \in C(\partial\Omega)$ and every point of $\partial\Omega$ is regular.

To prove this, we will first require the following lemma.

Lemma 7.3. *Let g be continuous at the regular point $z \in \partial\Omega$. Then, for every $\varepsilon > 0$, there exists $w \in \mathcal{S}_g$ such that $w(z) \geq g(z) - \varepsilon$.*

Proof. Let $\varepsilon > 0$ be given and choose $\delta > 0$ such that

$$|g(x) - g(z)| < \varepsilon, \quad \forall x \in B_\delta(z) \cap \partial\Omega.$$

Next, let $M > 0$ be so large that

$$M\varphi(x) + 2\|g\|_\infty < 0 \quad \text{on } \partial\Omega \setminus B_\delta(x);$$

where φ is a barrier at z . Clearly, the function

$$w(x) := M\varphi(x) + g(z) - \varepsilon$$

is continuous on $\bar{\Omega}$ and subharmonic in Ω . Moreover, a direct calculation verifies that

$$w(z) = g(z) - \varepsilon.$$

Thus, we need only check that $w \leq g$ on $\partial\Omega$. If $x \in B_\delta(x) \cap \partial\Omega$ then

$$w(x) = M\varphi(x) + g(z) - \varepsilon \leq g(z) - \varepsilon < g(x)$$

and we are done. Otherwise, if $x \in \partial\Omega \setminus B_\delta(x)$ then

$$w(x) < M\varphi(x) + g(z) - 2\|g\|_\infty + g(z) \leq g(x),$$

where the last inequality follows from the fact that $g(z) - g(x) \leq 2\|g\|_\infty$. $\square$

Proof of Theorem. We are now prepared to prove the theorem. Given $\varepsilon > 0$, the lemma allows us to choose a function $w \in \mathcal{S}_g$ such that

$$w(z) \geq g(z) - \varepsilon.$$

Since $w(z) \in \mathcal{S}_g$, we have $w \leq u$ on Ω . Therefore,

$$\liminf_{\Omega \ni x \rightarrow z} u(x) \geq \liminf_{\Omega \ni x \rightarrow z} w(x) = w(z) \geq g(z) - \varepsilon.$$

Letting $\varepsilon \rightarrow 0$, the above becomes

$$\liminf_{\Omega \ni x \rightarrow z} u(x) \geq g(z).$$

Conversely, given $\varepsilon > 0$, we choose $w \in \mathcal{S}_{-g}$ such that

$$w(z) \geq -g(z) - \varepsilon.$$

Let now $v \in \mathcal{S}_g$ be given; clearly $v + w \in C(\bar{\Omega}) \cap \mathfrak{Sub}(\Omega)$ satisfies $v + w \leq 0$ on $\partial\Omega$. Thus, an application of the strong maximum principle forces $v \leq -w$ in Ω . By definition, we must then have $u \leq -w$ in Ω . Thus,

$$\limsup_{\Omega \ni x \rightarrow z} u(x) \leq \limsup_{\Omega \ni x \rightarrow z} -w(x) = -w(z) \leq g(z) + \varepsilon.$$

Sending $\varepsilon \rightarrow 0$, we get $\limsup_{\Omega \ni x \rightarrow z} u(x) \leq g(z)$. $\square$

7.3. A Sufficient Regularity Condition. The abstract definition of a regular point and a barrier can often make it difficult to tell whether a given Dirichlet problem actually admits a solution according to the Perron method. Luckily, there exists a simple method that can often be used to show that a point is regular.

Proposition 7.4. *Let Ω be a bounded domain, fix $z \in \partial\Omega$, and suppose that there exists $r > 0$ and $y \in \mathbb{R}^n$ such that*

$$B(y, r) \cap \Omega = \emptyset \quad \text{and} \quad \overline{B(y, r)} = \{z\}.$$

Then, z is a regular point.

Proof. Suppose for simplicity that $n \geq 3$. We need only construct a barrier at z . Define a function

$$\varphi(x) := \frac{1}{|x - y|^{n-2}} - \frac{1}{r^{n-2}}, \quad \text{for } x \in \overline{\Omega}.$$

Clearly, φ is harmonic in $\mathbb{R}^n \setminus \{y\}$. Also, having $z \in \overline{B(y, r)}$ implies $z \in \partial B(y, r)$ whence $\varphi(z) = 0$. If $x \in \partial\Omega \setminus \{z\}$ then we must have $|x - y| > r$ so that $\varphi(x) < 0$. $\square$

We conclude with a variant of the maximum principle, which we state without proof.

Theorem 7.5. *Let $\Omega \subseteq \mathbb{R}^n$ be a bounded domain and let $u \in C(\Omega)$ be subharmonic in Ω . If*

$$\limsup_{\substack{x \in \Omega \\ x \rightarrow z}} u(x) \leq 0$$

for each $z \in \partial\Omega$, then $u \leq 0$ in all of Ω .

8. A VARIATIONAL APPROACH TO THE DIRICHLET PROBLEM

Thus far we have only been able to guarantee the existence of solutions to the Dirichlet problem (3.1) when Ω is a ball. Although this gave way to regularity results, we would now like to address the same existence question for more general $\Omega \subseteq \mathbb{R}^n$. Ideally, we would like to have a similar existence result for *bounded and open* Ω . To achieve this, we will need to view the Dirichlet problem in a completely different light. More precisely, we will treat problem (3.1) as a variational problem in the setting of Sobolev spaces. To help motivate this perspective, we give the following lemma.

Lemma 8.1. *Let $\Omega \subseteq \mathbb{R}^n$ be an open set and $u \in C^2(\Omega)$. Then, u is harmonic in Ω if and only if*

$$\int_{\Omega} \nabla u \cdot \nabla \varphi \, dm = 0, \quad \forall \varphi \in C_c^\infty(\Omega).$$

Here, $C_c^\infty(\Omega)$ denotes the space of all $\mathbb{R}$ -valued smooth functions having compact support in Ω .

Proof. The proof is obvious because, after an integration by parts, one has

$$\int_{\Omega} \nabla u \cdot \nabla \varphi \, dm = - \int_{\Omega} \varphi \Delta u \, dm$$

for all $\varphi \in C_c^\infty(\Omega)$. □

We now consider the space $C^1(\bar{\Omega})$, which consists of all continuously differentiable functions $u : \bar{U} \rightarrow \mathbb{R}$. For any u belonging to this space, we can safely define the following quantity:

$$E(u) := \int_{\Omega} |\nabla u|^2. \quad (8.1)$$

The quantity $E(u)$ is called the *Dirichlet energy* of u , and E is its corresponding energy functional. If we assume that Ω is bounded, then $E(u)$ will be finite for all $u \in C^1(\bar{\Omega})$. Otherwise, we cannot guarantee anything.

Proposition 8.2. *Let Ω be an open subset of $\mathbb{R}^n$ and let $u \in C^2(\Omega)$ be such that $E(u) < \infty$. The following statements are equivalent:*

- (1) $\Delta u \equiv 0$ in Ω , i.e. u is harmonic in Ω ;
- (2) $E(u) \leq E(u + v)$ for all $v \in C_c^\infty(\Omega)$.

Thus, provided u is sufficiently regular, we can characterize the harmonicity of u by the functional $E(\cdot)$.

Proof. First assume that u is harmonic in Ω ; we will show that $E(u) \leq E(u + v)$ for every $v \in C_c^\infty(\Omega)$. To this end, fix a test function $v \in C_c^\infty(\Omega)$ and calculate

$$\begin{aligned} E(u + v) - E(u) &= \int_{\Omega} (\nabla u + \nabla v) \cdot (\nabla u + \nabla v) - \int_{\Omega} |\nabla u|^2 \\ &= \int_{\Omega} |\nabla u|^2 + 2 \int_{\Omega} \nabla u \cdot \nabla v + \int_{\Omega} |\nabla v|^2 - \int_{\Omega} |\nabla u|^2 \\ &= E(v) + 2 \int_{\Omega} \nabla u \cdot \nabla v. \end{aligned}$$

On the other hand, an integration by parts shows that

$$\int_{\Omega} \nabla u \cdot \nabla v = - \int_{\Omega} v \Delta u = 0,$$

by hypothesis on u . Therefore, $E(u + v) - E(u) = E(v) \geq 0$. Conversely, suppose that $E(u) \leq E(u + v)$ for all $v \in C_c^\infty(\Omega)$. Let $\varepsilon \in \mathbb{R}$ be given and compute

$$\begin{aligned} E(u + \varepsilon v) &= \int_{\Omega} (\nabla u + \varepsilon \nabla v) \cdot (\nabla u + \varepsilon \nabla v) \\ &= E(u) + \varepsilon^2 E(v) + 2\varepsilon \int_{\Omega} \nabla u \cdot \nabla v. \end{aligned}$$

Hence, $E(u + \varepsilon v) - E(u) \geq 0$ means that

$$-2\varepsilon \int_{\Omega} \nabla u \cdot \nabla v \leq \varepsilon^2 E(v), \quad \forall v \in C_c^\infty(\Omega), \varepsilon \in \mathbb{R}.$$

Let now $\varepsilon > 0$; the above says that

$$\int_{\Omega} \nabla u \cdot \nabla v \geq -\varepsilon E(v).$$

On the other hand, replacing ε by $-\varepsilon$, we obtain

$$2\varepsilon \int_{\Omega} \nabla u \cdot \nabla v \leq \varepsilon^2 E(v)$$

whence

$$2 \int_{\Omega} \nabla u \cdot \nabla v \leq \varepsilon E(v).$$

Combining these inequalities shows that

$$\left| \int_{\Omega} \nabla u \cdot \nabla v \right| \leq \frac{\varepsilon}{2} E(v), \quad \forall \varepsilon > 0.$$

After an integration by parts, it follows that $\int_{\Omega} v \Delta u = 0$ for every $v \in C_c^\infty(\Omega)$. Hence, u is harmonic in Ω . $\square$

What exactly does this proposition tell us? Well, put informally, it says that finding a harmonic function $u \in \Omega$ is equivalent to minimizing the energy functional $E(\cdot)$ over the set

$$\{u \in C^2(\Omega) : E(u) < \infty\}.$$

However, this will not help us much when it comes to existence. Indeed, this criterion is only applicable if we already have a candidate function $u \in C^2(\Omega)$, with $E(u) < \infty$, that we suspect to be harmonic. Rather than applying this proposition directly, we will extrapolate from the proof and try to determine which properties will be key when determining if there exist solutions to the problem (3.1).

8.1. Minimizing the Dirichlet Energy in $H^1(\Omega)$. We first work to significantly generalize the Dirichlet problem. The idea is as follows: it is very hard to directly search for a smooth solution to a given PDE. Instead, we would like to use the sophisticated results from functional analysis to find a *weak solution* to a PDE (which is a much easier problem). After having found such a solution, one often tries to show that the solution is sufficiently regular, (e.g. C^2) at which point one would have actually solved the original (classical) PDE.

First things first, we have to somehow develop a weak formulation of the PDE

$$\Delta u \equiv 0 \quad \text{in } \Omega.$$

For simplicity, let us first assume that $u \in C^2(\Omega)$. If u is harmonic in Ω , then for every $\varphi \in C_c^\infty(\Omega)$:

$$-\int_{\Omega} \varphi \Delta u = 0.$$

After an integration by parts, we find that

$$\int_{\Omega} \nabla u \cdot \nabla \varphi = 0, \quad \forall \varphi \in C_c^\infty(\Omega). \quad (8.2)$$

Conversely, suppose that the above is satisfied for some $u \in C^2(\Omega)$. Integration by parts yields

$$\int_{\Omega} \varphi \Delta u = 0, \quad \forall \varphi \in C_c^\infty(\Omega).$$

Consequently, we must then have $\Delta u \equiv 0$ in Ω . Since (8.2) makes sense even for $u \in H^1(\Omega)$, we have a natural generalization of harmonicity.

Definition 5. A function $u \in H^1(\Omega)$ is said to be *weakly harmonic* in Ω if it satisfies equation (8.2).

Equipped with this definition, we know what it means for a function $u \in H^1(\Omega)$ to be weakly harmonic. Recall that $H_0^1(\Omega)$ is the closure of $C_c^\infty(\Omega)$ in $H^1(\Omega)$. In particular, $H_0^1(\Omega)$ is a Hilbert space.

To achieve any kind of existence, we will first require the following important inequality due to Friedrich:

Theorem 8.3 (Friedrich's Inequality). *Let Ω be a bounded open subset of $\mathbb{R}^n$. There exists a constant $C > 0$ such that*

$$\|u\|_{L^2(\Omega)} \leq C \|\nabla u\|_{L^2(\Omega)}$$

for all $u \in H_0^1(\Omega)$.

Proof. By density, it is enough to prove the inequality for $u \in C_c^\infty(\Omega)$. Without loss of generality, assume that $\Omega \subseteq (0, \alpha)^n$ for some $\alpha > 0$. Fixing a point $x = (x', x_n)$ in Ω , we have

$$\begin{aligned} |u(x)| &= \left| \int_0^{x_n} \partial_{x_n} u(x', t) dt \right| \leq \int_0^\alpha |\partial_{x_n} u(x', t)| dt \\ &\leq \alpha^{1/2} \|\partial_{x_n} u(x', \cdot)\|_{L^2(0, \alpha)}. \end{aligned}$$

Squaring both sides yields

$$|u(x)|^2 \leq \alpha \int_0^\alpha |\partial_{x_n} u(x', t)|^2 dt$$

where, upon integrating over $x' = \text{constant}$, we find that

$$\int_0^\alpha |u(x', t)|^2 dt \leq \alpha^2 \int_0^\alpha |\partial_{x_n} u(x', t)|^2 dt.$$

Finally, we integrate over $x' \in (0, \alpha)^{n-1}$ to get

$$\begin{aligned} \int_\Omega |u|^2 &= \int_{(0, \alpha)^n} |u|^2 \leq \alpha^2 \int_{(0, \alpha)^n} |\partial_{x_n} u|^2 \\ &\leq \alpha^2 \int_\Omega |\nabla u|^2. \end{aligned}$$

Taking square roots on both sides yields the desired inequality. $\square$

Corollary 8.4. *Let $\Omega \subseteq \mathbb{R}^n$ be open and bounded. There exists a constant $C > 0$ such that*

$$\|u\|_{H^1(\Omega)} \leq C \|\nabla u\|_{L^2(\Omega)}$$

for all $u \in H_0^1(\Omega)$. In particular, $\|\nabla u\|_{L^2(\Omega)}$ is an equivalent norm for the topology of $H_0^1(\Omega)$.

As the completion of $C_c^\infty(\Omega)$ is $H_0^1(\Omega)$, it makes sense intuitively to treat functions in $H_0^1(\Omega)$ as those H^1 -functions vanishing on $\partial\Omega$. With this perspective in mind, it make sense to give a weak formulation of (3.1).

Definition 6. Given a function $g \in H^1(\Omega)$, we will say a function $u \in H^1(\Omega)$ solves (3.1) *weakly* if u is weakly harmonic in Ω and $u - g \in H_0^1(\Omega)$.

Theorem 8.5. *Let Ω be open and bounded, suppose also that $g \in H^1(\Omega)$. There exists a function $u \in H^1(\Omega)$ such that*

- (1) u is weakly harmonic in Ω , i.e. (8.2) holds in Ω ;
- (2) $u - g \in H_0^1(\Omega)$.

Moreover, this function u is unique modulo sets of Lebesgue measure zero.

Proof. Let us first show that such a function u exists. For this, we will make use of the Dirichlet energy function $E(\cdot)$ defined earlier. Clearly, $E(\cdot)$ makes sense as a function $H^1(\Omega) \rightarrow \mathbb{R}$. Consider now the space

$$\mathcal{A} := \{g + v : v \in H_0^1(\Omega)\} \subseteq H^1(\Omega).$$

Since $0 \in H_0^1(\Omega)$, it is obvious that $g \in \mathcal{A}$. In fact, because $g \in H^1(\Omega)$, we automatically know that $E(g) = \int_{\Omega} |\nabla g|^2 < \infty$. Hence,

$$\mu := \inf_{u \in \mathcal{A}} E(u)$$

is non-negative and finite. Pick now a sequence $\{u_k\}_{k \in \mathbb{N}}$ in $\mathcal{A}$ such that

$$E(u_k) \xrightarrow{k \rightarrow \infty} \mu.$$

Ideally, we would like to show that there is a function $u \in H^1(\Omega)$ which minimizes $E(\cdot)$ over $\mathcal{A}$, i.e. that $E(u) = \mu$. From there, given our previous criterion for harmonicity, it would be reasonable to expect u to be weakly harmonic in Ω . Now, for indices $k, j \geq 1$ we calculate

$$\begin{aligned} E(u_k - u_j) &= 2E(u_k) + 2E(u_j) - 4E\left(\frac{u_k + u_j}{2}\right) \\ &\leq 2E(u_k) + 2E(u_j) - 4\mu. \end{aligned}$$

This last inequality is valid because

$$\frac{u_k + u_j}{2} \in \mathcal{A}.$$

Since $E(u_k), E(u_j) \rightarrow \mu$ as $k, j \rightarrow \infty$, we see that $E(u_k - u_j) \rightarrow 0$ as $k, j \rightarrow \infty$. Writing this out explicitly says that

$$\int_{\Omega} |\nabla u_k - \nabla u_j|^2 \xrightarrow{j, k \rightarrow \infty} 0.$$

In other words, $\{\nabla u_k\}_{k \in \mathbb{N}}$ is Cauchy in $L^2(\Omega)$. By Friedrich's inequality, it follows that $\{u_k\}_{k \in \mathbb{N}}$ is Cauchy in $H^1(\Omega)$. Thus, we can find $u \in H^1(\Omega)$ to which u_k converges. Provided we can show that E is continuous on $H^1(\Omega)$, it would follow that

$$E(u) = \lim_{k \rightarrow \infty} E(u_k) = \mu.$$

To this end, let us fix $w \in H^1(\Omega)$ and pick a sequence $\{w_k\}_{k \in \mathbb{N}}$ in $H^1(\Omega)$ converging to w with k . A standard estimate gives

$$\begin{aligned} |E(w) - E(w_k)| &= \left| \int_{\Omega} (\nabla w \cdot \nabla w - \nabla w_k \cdot \nabla w_k) \right| \\ &= \left| \int_{\Omega} (\nabla w \cdot \nabla w - \nabla w \cdot \nabla w_k + \nabla w \cdot \nabla w_k - \nabla w_k \cdot \nabla w_k) \right| \\ &\leq \int_{\Omega} |\nabla w| |\nabla w - \nabla w_k| + \int_{\Omega} |\nabla w_k| |\nabla w - \nabla w_k| \\ &\leq \|w\|_{H^1(\Omega)} \|w - w_k\|_{H^1(\Omega)} + M \|w - w_k\|_{H^1(\Omega)}. \end{aligned}$$

Thus, E is a continuous functional. By our earlier remarks, we see that $E(u) = \mu$.

Next, we argue that $u - g$ belongs to $H_0^1(\Omega)$. For this, note that $u_k \in \mathcal{A}$ implies $u_k - g \in H_0^1(\Omega)$ for every index k . Since $H_0^1(\Omega)$ is a closed subspace of $H^1(\Omega)$ and

$$u_k - g \rightarrow u - g,$$

it follows that $u - g \in H_0^1(\Omega)$. With this, we show that u is weakly harmonic in Ω . Namely, we fix a test function $\varphi \in C_c^\infty(\Omega)$ and prove that

$$\int_{\Omega} \nabla u \cdot \nabla \varphi = 0.$$

To this end, let us fix $\varepsilon \in \mathbb{R}$. Then, $u + \varepsilon\varphi$ is an element of $\mathcal{A}$ because

$$u + \varepsilon\varphi = g + \underbrace{(u - g + \varepsilon\varphi)}_{\in H_0^1(\Omega)}$$

whence $E(u) \leq E(u + \varepsilon\varphi)$. Expanding $E(u + \varepsilon\varphi)$ gives

$$\begin{aligned} E(u) &\leq \int_{\Omega} (\nabla u + \varepsilon \nabla \varphi) \cdot (\nabla u + \varepsilon \nabla \varphi) \\ &= E(u) + \varepsilon^2 E(\varphi) + 2\varepsilon \int_{\Omega} \nabla u \cdot \nabla \varphi. \end{aligned}$$

Thus,

$$-2\varepsilon \int_{\Omega} \nabla u \cdot \nabla \varphi \leq \varepsilon^2 E(\varphi)$$

for all $\varepsilon \in \mathbb{R}$. As in the proof of Proposition 8.2, this forces

$$\int_{\Omega} \nabla u \cdot \nabla \varphi = 0.$$

As $\varphi \in C_c^\infty(\Omega)$ was arbitrary, this shows that $\Delta u = 0$ weakly in Ω . With this, all that remains is the uniqueness of u . Suppose that $\tilde{u}$ is any element of $H^1(\Omega)$

satisfying the two properties in the statement of our theorem. Then,

$$u - \tilde{u} = (u - g) - (\tilde{u} - g) \in H_0^1(\Omega)$$

so that

$$\int_{\Omega} |\nabla u - \nabla \tilde{u}|^2 = \int_{\Omega} (\nabla u - \nabla \tilde{u}) \cdot (\nabla u - \nabla \tilde{u}) \quad (8.3)$$

$$= \int_{\Omega} \nabla u \cdot \nabla (u - \tilde{u}) - \int_{\Omega} \nabla u \cdot \nabla (u - \tilde{u}). \quad (8.4)$$

Since $\int_{\Omega} \nabla u \cdot \nabla \varphi$ vanishes for all $\varphi \in C_c^\infty(\Omega)$, a standard density argument shows that we in fact have this for all $\varphi \in H_0^1(\Omega)$. Similarly,

$$\int_{\Omega} \nabla \tilde{u} \cdot \nabla \varphi = 0, \quad \forall \varphi \in H_0^1(\Omega).$$

It then follows from (8.4) that

$$\int_{\Omega} |\nabla u - \nabla \tilde{u}|^2 = 0.$$

In particular, $\|\nabla(u - \tilde{u})\|_{L^2(\Omega)} = 0$. By Friedrich's inequality, we actually get $\tilde{u} = u$ in $H^1(\Omega)$. With this, the proof is complete. $\square$

9. REGULARITY OF WEAKLY HARMONIC FUNCTIONS

We now ask our selves what being weakly harmonic in an open set Ω says about the differentiability of the function $u \in H^1(\Omega)$. If $u \in C^2(\Omega)$ is harmonic in Ω , then we know that u is smooth in Ω . In this section, we actually show that weakly harmonic functions agree (except possibly on a set of Lebesgue measure zero) with a harmonic function in Ω . Thus, all weakly harmonic functions are classically harmonic and hence smooth.

Theorem 9.1. *Let $u \in H^1(\Omega)$ be weakly harmonic, i.e. suppose that*

$$\int_{\Omega} \nabla u \cdot \nabla \varphi = 0, \quad \forall \varphi \in C_c^\infty(\Omega).$$

Then, up to modification on a set of measure zero, we have $u \in C^\infty(\Omega)$.

Proof. Let $\{K_n\}_{n \in \mathbb{N}}$ be a compact exhaustion of Ω . By definition, we will have $K_n \nearrow \Omega$. Fix now a natural number n and consider an arbitrary compact set A contained in the interior of K_n . For every $\varepsilon > 0$, let u_ε denote the mollified version of u . If $\varepsilon > 0$ is sufficiently small, we know that u_ε will be smooth in the interior of K_n . In fact, we can find $\delta > 0$ such that

$$\overline{B(x, \varepsilon)} \subseteq \overline{B(x, \delta)} \subseteq \text{Int}(K_n)$$

for all $x \in \Lambda$ and all $0 < \varepsilon \leq \delta$. If $\varepsilon > 0$ is sufficiently small and $x \in \text{Int}(K_n)$ is given, a standard computation shows that

$$\begin{aligned} \Delta u_\varepsilon(x) &= \int_{\Omega} \Delta_x \eta_\varepsilon(x-y) u(y) \, dy = \int_{\Omega} \Delta_y \eta_\varepsilon(x-y) u(y) \, dy \\ &= - \int_{\Omega} \nabla_y \eta_\varepsilon(x-y) \cdot \nabla_y u(y) \, dy = 0. \end{aligned}$$

In this last step, we have used that $\eta_\varepsilon(x-\cdot)$ is a smooth function with compact support in Ω , for all $\varepsilon > 0$ small. Of course, η_ε denotes our mollifier. In any case, we see that u_ε is harmonic in the interior of K_n for all $\varepsilon > 0$ sufficiently small. Given a point $x \in \Lambda$ and $\varepsilon, \gamma > 0$ small, the mean value formula tells us that

$$|u_\varepsilon(x) - u_\gamma(x)| = \left| \int_{B(x,\delta)} (u_\varepsilon - u_\gamma) \, dm \right| \leq \int_{B(x,\delta)} |u_\varepsilon - u_\gamma| \, dm.$$

Because $u_\varepsilon \rightarrow u$ in $L^1_{\text{loc}}(\Omega)$, the above shows that u_ε is Cauchy in the uniform norm on $\Lambda \subset \Omega$. Consequently, u_ε converges locally uniformly on the interior of K_n . By Harnack's theorem, u_ε converges to a function $w \in C^\infty(\text{Int}(K_n))$ which is harmonic in the interior of K_n . Since u_ε converges to u in $L^1(K_n)$, it follows that $u = w_n$ almost everywhere in the interior of K_n . Since this works for all $n \geq 1$ and $K_n \nearrow \Omega$ is an exhaustion of Ω , the claim follows. $\square$

9.1. Boundary Regularity of Solutions. This subsection is figurative at best, and only serves to illustrate the notion of a trace map when the domain Ω is nice.

Proposition 9.2. *Let $\Omega = (0, 1)$ be the open unit interval and let $u \in H_0^1(\Omega)$. Then, we can find $v \in C(\bar{\Omega})$ with $v = u$ almost everywhere and $v(0) = v(1) = 0$.*

Proof. Select a sequence $\{v_k\}_k$ in $C_c^\infty(\Omega)$ that converges to u in $H^1(\Omega)$ as $k \rightarrow \infty$. Testing with $x \in \Omega$ and $v \in C_c^\infty(\Omega)$, we find that

$$|v(x)| \leq \int_0^x |v'| \leq \sqrt{x} \|v'\|_{L^2(\Omega)}.$$

In particular, the sequence $\{v_k\}$ is Cauchy in the uniform norm of $(0, 1)$. Thus, there is some $w \in C(\Omega)$ such that $v_k \rightarrow w$ uniformly on I . Since $v_k \rightarrow u$ in $L^2(\Omega)$, we must have $w = u$ almost everywhere on Ω . If we can prove that

$$\lim_{x \rightarrow 0} w(x) = \lim_{x \rightarrow 1} w(x) = 0$$

then we would be done. Passing to the limit in the inequality with $v(x)$, we see that

$$|w(x)| \leq \sqrt{x} \|u'\|_{L^2(\Omega)}$$

for all $x \in \Omega$. Therefore, $w(x) \rightarrow 0$ as $x \rightarrow 0$. One treats the case $x \rightarrow 1$ similarly. $\square$

In two dimensions this is handled similarly.

Theorem 9.3. *Let $\Omega := (0, 1) \times (0, 1) \subset \mathbb{R}^2$ and let $h > 0$. Define*

$$\mathsf{T}_h : C_c^\infty(\Omega) \rightarrow L^2((0, 1))$$

via the rule $(\mathsf{T}_h \varphi)(x) = \varphi(x, h)$. Then, T_h uniquely extends to a bounded linear map

$$H_0^1(\Omega) \rightarrow L^2((0, 1)).$$

Moreover, for each $u \in H_0^1(\Omega)$ there holds $\mathsf{T}_h u \rightarrow 0$ in $L^2((0, 1))$ as $h \rightarrow 0$.

Proof. For the first part, it is enough to show that T_h is bounded. If $\varphi \in C_c^\infty(\Omega)$, for any $x \in (0, 1)$:

$$|(\mathsf{T}_h \varphi)(x)| \leq \int_0^h |\partial_2 \varphi(x, x_2)| dx_2 \leq \sqrt{h} \|\partial_2 \varphi(x, \cdot)\|_{L^2(0, 1)}.$$

Squaring both sides and integrating over x , we find that

$$\|\mathsf{T}_h \varphi\|_{L^2(0, 1)}^2 \leq h \|\nabla \varphi\|_{L^2(\Omega)}^2 \leq h \|\varphi\|_{H^1(\Omega)}^2.$$

Thus, we can extend T_h to all of $H_0^1(\Omega)$, and moreover the bound above shall hold for all $\varphi \in H_0^1(\Omega)$. $\square$

Finally, we give a theorem due to Nirenberg.

Theorem 9.4. *Let $\Omega = (0, 1)^n$ and assume $u \in H_0^1(\Omega)$. If u is continuous at $z \in \partial\Omega$, then $u(z) = 0$.*

Proof. Without loss of generality, we may assume that $z = 0$. Let $h > 0$ be small and consider the open set

$$B := E \times (0, h) \subset \mathbb{R}^n,$$

where $E = (0, h^{n-1})^{n-1} \subset \mathbb{R}^{n-1}$. Thus, B has Lebesgue measure equal to h^2 . Fix $\varphi \in C_c^\infty(\Omega)$ and let $x = (x', x_n) \in B$ be given. We know that

$$|\varphi(x)| \leq \int_0^{x_n} |\partial_n \varphi(x', t)| dt \leq \int_0^h |\partial_n \varphi(x', t)| dt.$$

Squaring both sides and applying Hölder inequality shows that

$$|\varphi(x)|^2 \leq h \int_0^h |\partial_n \varphi(x', t)|^2 dt.$$

In particular, we find that

$$\int_0^h |\varphi(x', t)|^2 dt \leq h^2 \int_0^h |\partial_n \varphi(x', t)|^2 dt$$

with $x' \in E$ fixed. Finally, we integrate over $x' \in E$ to discover that

$$\int_B |\varphi|^2 \leq h^2 \int_B |\partial_n \varphi|^2 \leq h^2 \int_B |\nabla \varphi|^2.$$

This implies that

$$\int_B |\varphi|^2 \leq \int_B |\nabla \varphi|^2.$$

Since this inequality holds for all $\varphi \in C_c^\infty(\Omega)$, a density argument shows that the above holds for all $\varphi \in H_0^1(\Omega)$. Therefore,

$$\int_B |u|^2 \leq \int_B |\nabla u|^2.$$

Since u is continuous at 0 and ∇u is integrable on Ω , sending $h \rightarrow 0$ in the above yields

$$|u(0)|^2 \leq 0.$$

This completes the proof. $\square$

10. WEAK SOLUTIONS TO THE POISSON PROBLEM

Having discussed weak solutions to the Dirichlet problem (3.1), it is natural consider weak solutions to the homogenous Poisson problem, formulated below for *bounded* open Ω in $\mathbb{R}^n$:

$$\begin{cases} \Delta u + f \equiv 0 & \text{in } \Omega, \\ u \equiv 0 & \text{on } \partial\Omega. \end{cases} \quad (10.1)$$

Here, we will assume that $f \in L^2(\Omega)$. Arguing as in the previous section, we wish to find a function $u \in H_0^1(\Omega)$ such that

$$\int_\Omega \nabla u \cdot \nabla \varphi = \int_\Omega f \varphi, \quad \forall \varphi \in C_c^\infty(\Omega).$$

Note that the homogenous boundary condition is reflected in the requirement that $u \in H_0^1(\Omega)$. In the hopes of showing that a solution exists, we first consider the map

$$\Phi : H_0^1(\Omega) \ni \varphi \mapsto \int_\Omega f \varphi \in \mathbb{R}.$$

Clearly, Φ is linear. To see that Φ is also bounded, we invoke the Cauchy-Schwarz inequality as follows:

$$|\Phi(\varphi)| \leq \int_\Omega |f| |\varphi| \leq \|f\|_{L^2(\Omega)} \|\varphi\|_{L^2(\Omega)} \leq \|f\|_{L^2(\Omega)} \|\varphi\|_{H^1(\Omega)}.$$

Hence, Φ is an element of the dual space $(H_0^1(\Omega))^*$. Now, for $u, v \in H_0^1(\Omega)$ let us define

$$A(u, v) := \int_\Omega \nabla u \cdot \nabla v.$$

By the same reasoning, A is bounded as a map $H_0^1(\Omega) \times H_0^1(\Omega) \rightarrow \mathbb{R}$. By defining

$$\langle u, v \rangle := A(u, v),$$

we induce a symmetric bilinear form on $H_0^1(\Omega) \times H_0^1(\Omega)$. Clearly,

$$0 \leq \langle u, u \rangle = \int_{\Omega} |\nabla u|^2 \leq \|\nabla u\|_{L^2(\Omega)}^2 \leq \|u\|_{H^1(\Omega)}^2.$$

On the other hand, Friedrich's inequality tells us that

$$\langle u, u \rangle = \int_{\Omega} |\nabla u|^2 \geq C \|u\|_{H^1(\Omega)}^2$$

for some constant $C > 0$ independent of $u \in H_0^1(\Omega)$. These facts combined show that $\langle \cdot, \cdot \rangle$ is an inner product on $H_0^1(\Omega)$ inducing an equivalent topology. Hence, by the Riesz representation theorem, we can find an element $u \in H_0^1(\Omega)$ with the property that

$$\int_{\Omega} f \varphi = \Phi(\varphi) = \langle u, \varphi \rangle = \int_{\Omega} \nabla u \cdot \nabla \varphi$$

for all $\varphi \in H_0^1(\Omega)$. In particular, we have

$$\int_{\Omega} f \varphi = \int_{\Omega} \nabla u \cdot \nabla \varphi, \quad \forall \varphi \in C_c^\infty(\Omega).$$

This means that (10.1) can be solved in bounded open $\Omega \subset \mathbb{R}^n$. Furthermore, such a solution is necessarily unique. If u and v in $H_0^1(\Omega)$ both satisfy the above, then it follows that

$$\int_{\Omega} \nabla u \cdot \nabla \varphi = \int_{\Omega} \nabla v \cdot \nabla \varphi$$

for all test functions $\varphi \in C_c^\infty(\Omega)$. Equivalently, $\int_{\Omega} \nabla(u - v) \cdot \nabla \varphi = 0$ for all $\varphi \in C_c^\infty(\Omega)$. By density, we see that this identity holds for all $\varphi \in H_0^1(\Omega)$. Taking $\varphi := u - v$, this becomes

$$\int_{\Omega} |\nabla(u - v)|^2 = 0.$$

By Friedrich's inequality, one then has $\|u - v\|_{H^1(\Omega)} = 0$. To summarize what we have just done, we state the following theorem.

Theorem 10.1. *Let Ω be an open and bounded subset of $\mathbb{R}^n$. For any function $f \in L^2(\Omega)$, the homogeneous Poisson problem (10.1) has a unique weak solution $u \in H_0^1(\Omega)$.*

10.1. Solving the Nonhomogenous Poisson Problem. As a next step, it is natural to attempt to lift our previous existence result to the nonhomogenous case. That is, we seek a weak solution to the following problem:

$$\begin{cases} \Delta u + f \equiv 0 & \text{in } \Omega, \\ u \equiv g & \text{on } \partial\Omega. \end{cases} \quad (10.2)$$

Again, we are assuming that Ω is a bounded open subset of $\mathbb{R}^n$, that $f \in L^2(\Omega)$, and $g \in H^1(\Omega)$. Now, a weak solution of

$$\Delta u + f \equiv 0 \quad \text{in } \Omega$$

would necessarily satisfy

$$\int_{\Omega} \nabla u \cdot \nabla \varphi = \int_{\Omega} f \varphi, \quad \forall \varphi \in C_c^\infty(\Omega).$$

However, it does not make sense to look for $u \in H_0^1(\Omega)$ since we are working in the nonhomogenous setting. Instead, we seek a function $u \in H^1(\Omega)$ with the property that $u - g \in H_0^1(\Omega)$. Intuitively, this means that

$$(u - g)|_{\partial\Omega} \equiv 0 \quad \text{in some weak sense.}$$

Theorem 10.2. *Let $\Omega \subset \mathbb{R}^n$ be open and bounded, fix $f \in L^2(\Omega)$ and let $g \in H^1(\Omega)$. There exists a unique $u \in H^1(\Omega)$ such that*

(1) *for every $\varphi \in C_c^\infty(\Omega)$ there holds*

$$\int_{\Omega} \nabla u \cdot \nabla \varphi = \int_{\Omega} f \varphi;$$

(2) $u - g \in H_0^1(\Omega)$.

In other words, there exists a unique weak solution $u \in H^1(\Omega)$ to the problem (10.2).

Proof. Let us consider the map

$$H_0^1(\Omega) \rightarrow \mathbb{R}, \quad \varphi \mapsto \int_{\Omega} f \varphi - \int_{\Omega} \nabla g \cdot \nabla \varphi.$$

It is not hard to see that this is a bounded linear functional, and hence belongs to the dual space of $H_0^1(\Omega)$. Arguing as in the proof of the previous theorem, we can find $w \in H_0^1(\Omega)$ with the property that

$$\int_{\Omega} \nabla w \cdot \nabla \varphi = \int_{\Omega} f \varphi - \int_{\Omega} \nabla g \cdot \nabla \varphi$$

for all $\varphi \in H_0^1(\Omega) \supset C_c^\infty(\Omega)$. This implies that

$$\int_{\Omega} \nabla (w + g) \cdot \nabla \varphi = \int_{\Omega} f \varphi, \quad \forall \varphi \in H_0^1(\Omega).$$

Put now $u := w + g \in H^1(\Omega)$. Then, the above states that

$$\int_{\Omega} \nabla u \cdot \nabla \varphi = \int_{\Omega} f \varphi, \quad \forall \varphi \in C_c^\infty(\Omega).$$

Moreover, it is clear by construction that $u - g \in H_0^1(\Omega)$. All that remains is the uniqueness part. Suppose that u and $\tilde{u}$ both satisfy the statement of the theorem. Then, we get that

$$u - \tilde{u} = (u - g) - (\tilde{u} - g) \in H_0^1(\Omega).$$

On the other hand, having

$$\int_{\Omega} \nabla u \cdot \nabla \varphi = \int_{\Omega} f \varphi = \int_{\Omega} \nabla \tilde{u} \cdot \nabla \varphi$$

forces (by density)

$$\int_{\Omega} \nabla (u - \tilde{u}) \cdot \nabla \varphi = 0, \quad \forall \varphi \in H_0^1(\Omega).$$

Taking $\varphi := (u - \tilde{u})$ and invoking Friedrich's inequality shows that $u = \tilde{u}$ almost everywhere on Ω . $\square$

11. t-POISSON PROBLEMS AND NEUMANN BOUNDARY CONDITIONS

Given an open set $\Omega \subset \mathbb{R}^n$ with non-empty boundary, it is natural to consider solutions to the PDE

$$\Delta u + f \equiv 0 \quad \text{in } \Omega$$

satisfying a boundary condition on the *derivatives* of u . For instance, one might be interested in the Neumann-Poisson problem:

$$\begin{cases} \Delta u + f \equiv 0 & \text{in } \Omega, \\ \partial_{\nu} u \equiv 0 & \text{on } \partial\Omega. \end{cases} \quad (11.1)$$

Henceforth, let us assume that $f \in L^2(\Omega)$. For the sake of simplicity, let V denote one of $H_0^1(\Omega)$ and $H^1(\Omega)$. To help understand the weak formulation of such an equation, let us consider the problem of finding u belonging to V such that

$$\int_{\Omega} (\nabla u \cdot \nabla \varphi + t u \varphi) = \int_{\Omega} f \varphi, \quad \forall \varphi \in H^1(\Omega). \quad (11.2)$$

Here, $t \in \mathbb{R}$ is some fixed quantity. However, whether or not we can guarantee the existence of solutions to the problem above *will* depend on what t is. For the time being, let us figure out which classical equation (11.2) corresponds to in Ω . If $u \in C^2(\Omega)$ satisfies (11.2) for all test functions φ , an integration by parts shows that

$$\int_{\Omega} f \varphi = \int_{\Omega} (\nabla u \cdot \nabla \varphi + t u \varphi) = \int_{\Omega} (t u - \Delta u) \varphi.$$

As $\varphi \in C_c^\infty(\Omega)$ was arbitrary, we conclude (provided f is continuous on Ω) that

$$-\Delta u + t u \equiv f \quad \text{in } \Omega.$$

Conversely, if $u \in C^2(\Omega)$ satisfies the above, then f must be continuous on Ω . Another integration by parts shows that for each $\varphi \in C_c^\infty(\Omega)$:

$$\int_{\Omega} f\varphi = \int_{\Omega} (tu - \Delta u)\varphi = \int_{\Omega} (\nabla u \cdot \nabla \varphi + tu\varphi).$$

In other words, we see that (11.2) is simply the weak formulation of the PDE

$$-\Delta u + tu \equiv f \quad \text{in } \Omega.$$

without any boundary conditions. Now, the boundary conditions imposed on the problem will depend on which space we take to be V . If $V = H_0^1(\Omega)$, then our problem reduces to finding a function $u \in H_0^1(\Omega)$ for which (11.2) holds. Clearly, this corresponds to the weak formulation of the Dirichlet-type problem:

$$\begin{cases} -\Delta u + tu \equiv f & \text{in } \Omega, \\ u \equiv 0 & \text{on } \partial\Omega. \end{cases} \quad (11.3)$$

But what if we are merely looking for $u \in H^1(\Omega)$ solving (11.2)? To which classical problem would this be a weak solution? Clearly, u only being in $H^1(\Omega)$ would not affect the PDE in Ω , i.e. u is still a weak solution to the equation

$$-\Delta u + tu \equiv f \quad \text{in } \Omega.$$

To understand the boundary condition imposed by $u \in H^1(\Omega)$, we examine the behaviour of solutions $u \in C^1(\bar{\Omega}) \cap C^2(\Omega)$ when Ω has a C^1 -boundary. Then, being a solution to (11.2) would imply that for every $\psi \in C^1(\bar{\Omega})$ there holds

$$\begin{aligned} \int_{\Omega} f\psi &= \int_{\Omega} (\nabla u \cdot \nabla \psi + tu\psi) \\ &= \int_{\partial\Omega} \psi \partial_{\nu} u - \int_{\Omega} \psi \Delta u + \int_{\Omega} tu\psi. \end{aligned}$$

Since $u \in C^2(\Omega)$, this together with our earlier remarks imply that

$$\int_{\partial\Omega} \psi \partial_{\nu} u = \int_{\Omega} (f + \Delta u - tu)\psi = 0 \quad (11.4)$$

for all $\psi \in C^1(\bar{\Omega})$. Thus, we conclude that

$$\int_{\partial\Omega} \psi \partial_{\nu} u = 0, \quad \forall \psi \in C^1(\bar{\Omega}).$$

By a density argument, we see that $\partial_{\nu} u$ vanishes on $\partial\Omega$. Therefore, (11.2) with $u \in H^1(\Omega)$ is the weak formulation of the following Neumann-type problem:

$$\begin{cases} -\Delta u + tu \equiv f & \text{in } \Omega, \\ \partial_{\nu} u \equiv 0 & \text{on } \partial\Omega. \end{cases} \quad (11.5)$$

11.1. Existence of Weak Solutions. Next, we consider the existence of weak solutions to the problems (11.3)-(11.5). Let us begin with the weak formulation of (11.5), which involves finding $u \in H^1(\Omega)$ satisfying (11.2) for a given $t \in \mathbb{R}$. Define a bilinear form

$$a(\cdot, \cdot) : H^1(\Omega) \times H^1(\Omega) \rightarrow \mathbb{R}, \quad (u, v) \mapsto \int_{\Omega} (\nabla u \cdot \nabla v + tuv) .$$

It is not hard to check that $a(\cdot, \cdot)$ is continuous on the product space $H^1(\Omega) \times H^1(\Omega)$. Given $u \in H^1(\Omega)$ it is readily seen that for all $t \geq 0$:

$$0 \leq a(u, u) = \int_{\Omega} (|\nabla u|^2 + tu^2) \leq \max\{1, t\} \|u\|_{H^1(\Omega)}^2 .$$

In fact, if $t > 0$, then $a(u, u) = 0$ if and only if $u = 0$. Moreover, when $t > 0$ we have

$$a(u, u) = \int_{\Omega} (|\nabla u|^2 + tu^2) \geq \min\{1, t\} \|u\|_{H^1(\Omega)}^2 .$$

Therefore, when $t > 0$, $a(\cdot, \cdot)$ is an inner product on $H^1(\Omega)$ inducing a norm equivalent to that of $H^1(\Omega)$. Since $f \in L^2(\Omega)$, the map

$$H^1(\Omega) \ni v \mapsto \int_{\Omega} fv \in \mathbb{R}$$

is a continuous linear functional belonging to the topological dual $(H^1(\Omega))^*$. Invoking the Riesz-Representation theorem, we see that there exists a unique vector $u \in H^1(\Omega)$ with the property that

$$a(u, v) = \int_{\Omega} fv, \quad \forall v \in H^1(\Omega) .$$

Especially, we have found $u \in H^1(\Omega)$ such that

$$\int_{\Omega} (\nabla u \cdot \nabla \varphi + tu\varphi) = \int_{\Omega} f\varphi$$

for all $\varphi \in C_c^\infty(\Omega) \subseteq H^1(\Omega)$. We summarize our findings in the following theorem.

Theorem 11.1. *Let Ω be a proper open subset of $\mathbb{R}^n$, $t > 0$, and $f \in L^2(\Omega)$. There exists a unique function $u \in H^1(\Omega)$ satisfying (11.2). That is, there exists a unique weak solution to the Poisson-Neumann problem (11.5).*

Let us try to repeat this procedure for the problem in (11.3). Now, we are looking for a function $u \in H_0^1(\Omega)$ such that

$$\int_{\Omega} (\nabla u \cdot \nabla \varphi + tu\varphi) = \int_{\Omega} f\varphi$$

for all $\varphi \in C_c^\infty(\Omega)$. Let us redefine our bilinear form a as follows:

$$a : H_0^1(\Omega) \times H_0^1(\Omega) \rightarrow \mathbb{R}, \quad a(u, v) := \int_{\Omega} (\nabla u \cdot \nabla v + tuv) .$$

If we assume that Ω is a *bounded open set*, Friedrich's inequality ensures the existence of a constant $C > 0$ with the property that

$$\|u\|_{H^1(\Omega)} \leq C \|\nabla u\|_{L^2(\Omega)}, \quad \forall u \in H_0^1(\Omega).$$

In fact, examining the proof of Friedrich's inequality shows that

$$\|u\|_{L^2(\Omega)}^2 \leq \text{diam}(\Omega)^2 \|\nabla u\|_{L^2(\Omega)}^2$$

for all $u \in H_0^1(\Omega)$. Let us now consider the quantity

$$d := \frac{1}{\text{diam}(\Omega)^2} > 0.$$

For all t and $u \in H_0^1(\Omega)$ it is easily seen that

$$\begin{aligned} a(u, u) &= \int_{\Omega} (|\nabla u|^2 + tu^2) \leq \max\{1, t\} \int_{\Omega} (|\nabla u|^2 + u^2) \\ &= \max\{1, t\} \|u\|_{H^1(\Omega)}^2 . \end{aligned}$$

If we restrict ourselves to $t > -d$, there also holds

$$\begin{aligned} \int_{\Omega} (|\nabla u|^2 + tu^2) &= \|\nabla u\|_{L^2(\Omega)}^2 + t \|u\|_{L^2(\Omega)}^2 \\ &\geq \begin{cases} \|\nabla u\|_{L^2(\Omega)}^2 & \text{if } t \geq 0, \\ \|\nabla u\|_{L^2(\Omega)}^2 + t \text{diam}(\Omega)^2 \|\nabla u\|_{L^2(\Omega)}^2 & \text{if } -d < t < 0. \end{cases} \end{aligned}$$

For the latter, note that we have $|t \text{diam}(\Omega)^2| < 1$. In any case, we can find constants $C_{1,2} = C_{1,2}(t) > 0$ such that

$$C_1 \|\nabla u\|_{L^2(\Omega)}^2 \leq a(u, u) \leq C_2 \|u\|_{H^1(\Omega)}^2 .$$

Thus, by Friedrich's inequality, $a(u, u) = 0$ implies $u = 0$ in $H_0^1(\Omega)$. Since the norm $\|\nabla \cdot\|_{L^2(\Omega)}$ is equivalent to that of $H_0^1(\Omega)$, the above shows that $a(\cdot, \cdot)$ is an inner product on $H_0^1(\Omega)$ inducing an equivalent norm. Because the mapping

$$H_0^1(\Omega) \ni v \mapsto \int_{\Omega} f v \in \mathbb{R}$$

is a continuous linear functional on $H_0^1(\Omega)$, the Riesz representation theorem guarantees the existence of a unique $u \in H_0^1(\Omega)$ with $a(u, v) = \int_{\Omega} f v$ for all $v \in H_0^1(\Omega)$. Obviously, this implies the following theorem.

Theorem 11.2. *Let $\Omega \subset \mathbb{R}^n$ be a bounded open set and let $f \in L^2(\Omega)$. For any*

$$t > -\frac{1}{\text{diam}(\Omega)^2},$$

there exists a unique function $u \in H_0^1(\Omega)$ satisfying (11.2). That is, there exists a unique weak solution to the Dirichlet-type problem (11.3).

Having established this, we can easily find weak solutions to the Poisson type problem

$$\begin{cases} -\Delta u + tu \equiv f & \text{in } \Omega, \\ u \equiv g & \text{on } \partial\Omega \end{cases} \quad (11.6)$$

provided we are in the setting of the previous theorem. We make this precise below.

Theorem 11.3. *Let $\Omega \subset \mathbb{R}^n$ be bounded and open. Suppose that*

$$t > -\frac{1}{\text{diam}(\Omega)^2},$$

$f \in L^2(\Omega)$, and $g \in H^1(\Omega)$. Then, there exists a unique function $u \in H^1(\Omega)$ such that

$$\int_{\Omega} (\nabla u \cdot \nabla \varphi + tu\varphi) = \int_{\Omega} f\varphi, \quad \forall \varphi \in C_c^\infty(\Omega)$$

and $u - g \in H_0^1(\Omega)$. That is, there exists a unique weak solution to (11.6).

Proof. By the previous arguments, for any such t , the bilinear form

$$a : H_0^1(\Omega) \times H_0^1(\Omega) \rightarrow \mathbb{R}, \quad a(u, v) := \int_{\Omega} (\nabla u \cdot \nabla v + tuv)$$

is an inner product for the topology on $H_0^1(\Omega)$. Since the map

$$\ell : H_0^1(\Omega) \rightarrow \mathbb{R}, \quad v \mapsto \int_{\Omega} fv - \int_{\Omega} \nabla g \cdot \nabla v - \int_{\Omega} tgv$$

is a continuous linear functional, the Riesz representation theorem ensures the existence of a unique $u \in H_0^1(\Omega)$ such that $a(u, v) = \ell(v)$, for all $v \in H_0^1(\Omega)$. That is, so that

$$\int_{\Omega} ((\nabla u + \nabla g) \cdot \nabla v + t(u + g)v) = \int_{\Omega} fv, \quad \forall v \in H_0^1(\Omega).$$

In particular, this equality holds for all $v \in C_c^\infty(\Omega)$. If we set $w := u + g \in H^1(\Omega)$, it follows that

$$\int_{\Omega} (\nabla w \cdot \nabla \varphi + tw\varphi) = \int_{\Omega} fv, \quad \forall v \in H_0^1(\Omega).$$

Since $w - g = u \in H_0^1(\Omega)$, we are done. □

12. THE POINCARÉ INEQUALITY

In this brief section, we strengthen Friedrich's inequality. Namely, given an open and bounded subset Ω of $\mathbb{R}^n$, we show that there exists a constant $C > 0$ such that

$$\|u\|_{L^p(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)}$$

for all $u \in W_0^{1,p}(\Omega)$ and all $1 \leq p < \infty$. Here, $W_0^{1,p}(\Omega)$ is the closure of $C_c^\infty(\Omega)$ in $W^{1,p}(\Omega)$. Clearly, this inequality shows that the norm $\|\nabla(\cdot)\|_{L^p(\Omega)}$ is equivalent to the $W^{1,p}$ -norm on $W_0^{1,p}(\Omega)$.

Theorem 12.1 (Poincaré Inequality). *Let $1 \leq p < \infty$ and suppose $\Omega \subset \mathbb{R}^n$ is open and bounded. There exists a constant $C > 0$ such that*

$$\|u\|_{L^p(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)}$$

for all $u \in W_0^{1,p}(\Omega)$.

Proof. By their density in the Sobolev space $W_0^{1,p}(\Omega)$, it is enough to prove the claim for $u \in C_c^\infty(\Omega)$. Without loss of generality, we may assume there is $a > 0$ be such that $\Omega \subset (0, a)^n$. Let now $x = (x', x_n) \in \Omega$ be fixed and write

$$|u(x)| \leq \int_0^{x_n} |u_{x_n}(x', t)| dt \leq \int_0^a |u_{x_n}(t)| dt.$$

Next, we raise both sides of this expression to the power of p and apply Hölder's inequality

$$|u(x)|^p \leq \left(\int_0^a |u_{x_n}(t)| dt \right)^p \leq a^{p/q} \int_0^a |u_{x_n}(x', t)|^p dt.$$

Here, q is the conjugate exponent to p . Integrating over $x' = \text{constant}$ shows that

$$\int_0^a |u(x', t)|^p dt \leq a^{p/q+1} \int_0^a |u_{x_n}(x', t)|^p dt.$$

Finally, we integrate over $x' \in (0, a)^{n-1}$, thereby obtaining

$$\begin{aligned} \int_\Omega |u|^p &= \int_{(0,a)^n} |u(x)|^p dx \leq a^{p/q+1} \int_{(0,a)^n} |u_{x_n}(x)|^p dx \\ &= a^{p/q+1} \int_\Omega |u_{x_n}|^p \\ &\leq a^{p/q+1} \int_\Omega |\nabla u|^p. \end{aligned}$$

This completes the proof. □

Remark 12.1. In the context of the previous theorem, denote by $d > 0$ the diameter of Ω . Then, the proof tells us that

$$\|u\|_{L^p(\Omega)}^p \leq a^{p/q+1} \|\nabla u\|_{L^p(\Omega)}^p$$

for all $u \in W_0^{1,p}(\Omega)$. Taking p^{th} roots on either side shows that the constant C in the statement of Poincaré's inequality can be taken as

$$C := d^{\frac{1}{p} + \frac{1}{q}} = d = \text{diam}(\Omega).$$

13. THE SPECTRAL THEOREM FOR COMPACT SELF-ADJOINT OPERATORS

In this section, we shall prove the spectral theorem for compact self-adjoint linear operators on a Hilbert space $\mathcal{H}$. This result will prove to be of considerable use in the section that follows. Assuming basic results from operator theory, we shall be able to give a complete and self-contained proof of this powerful result.

In this section, we denote by $\mathbb{K}$ the ground field of a Hilbert space $\mathcal{H}$. Thus, $\mathbb{K}$ stands for one of either $\mathbb{R}$ or $\mathbb{C}$. For $\lambda \in \mathbb{K}$, the quantity $\bar{\lambda}$ represents the complex conjugate of λ . In this case where $\mathbb{K} = \mathbb{R}$, this is redundant but not harmful to the proof. Unless stated otherwise, T represents an arbitrary operator $\mathcal{H} \rightarrow \mathcal{H}$.

Definition 7. An operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is said to be *compact* if, for every bounded sequence (f_n) in $\mathcal{H}$, there is a subsequence (f_{n_k}) such that $(T(f_{n_k}))$ is convergent.

It is automatic that compact linear operators are bounded. To see this, suppose that a linear operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is compact but not bounded. Then, it follows that

$$\sup_{\substack{f \in \mathcal{H} \\ \|f\|=1}} \|T(f)\| = \infty.$$

Consequently, we can select a sequence of unit vectors (f_n) in $\mathcal{H}$ such that

$$\|T(f_n)\| \rightarrow \infty, \quad \text{as } n \rightarrow \infty.$$

In particular, for any subsequence (f_{n_k}) of (f_n) , we have $\lim \|T(f_{n_k})\| = \infty$. Since (f_n) is a bounded sequence in $\mathcal{H}$, this contradicts the compactness of the operator T . We conclude that T must be bounded and continuous.

Definition 8. Let $\langle \cdot, \cdot \rangle$ denote the inner product on $\mathcal{H}$. A linear operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is said to be *self-adjoint* (of Hermitian) if

$$\langle Tf, g \rangle = \langle f, Tg \rangle$$

for all $f, g \in \mathcal{H}$.

As a first step in the proof, we require an elementary lemma regarding the eigenvalues of a self-adjoint operator.

Lemma 13.1. *Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be a self-adjoint operator. Then, any eigenvalue of T is real.*

Proof. Let $\lambda \in \mathbb{C}$ be an eigenvalue of T and let $\varphi \neq 0$ be a corresponding eigenvector. Next, we calculate

$$\begin{aligned} \lambda \langle \varphi, \varphi \rangle &= \langle \lambda \varphi, \varphi \rangle = \langle T\varphi, \varphi \rangle = \langle \varphi, T\varphi \rangle \\ &= \langle \varphi, \lambda \varphi \rangle \\ &= \bar{\lambda} \langle \varphi, \varphi \rangle. \end{aligned}$$

Since φ is non-zero, it follows that $\lambda = \bar{\lambda}$. This is only possible if $\Im \lambda = 0$. $\square$

This was easy enough. In a similar vein, we shall now check that eigenvectors for distinct eigenvalues are orthogonal.

Lemma 13.2. *Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be a self-adjoint operator and let λ_1, λ_2 be distinct eigenvalues for T . If φ_1 and φ_2 are eigenvectors corresponding to λ_1 and λ_2 , respectively, then $\varphi_1 \perp \varphi_2$.*

Proof. We will show that $\langle \varphi_1, \varphi_2 \rangle = 0$. To this end, we compute

$$\begin{aligned} \lambda_1 \langle \varphi_1, \varphi_2 \rangle &= \langle \lambda \varphi_1, \varphi_2 \rangle = \langle T\varphi_1, \varphi_2 \rangle \\ &= \langle \varphi_1, T\varphi_2 \rangle \\ &= \langle \varphi_1, \lambda_2 \varphi_2 \rangle \\ &= \lambda_2 \langle \varphi_1, \varphi_2 \rangle. \end{aligned}$$

In this last step, we have applied the previous lemma to ensure $\lambda_2 \in \mathbb{R}$. Since $\lambda_1 \neq \lambda_2$ by assumption, the only possibility is to have $\langle \varphi_1, \varphi_2 \rangle = 0$. $\square$

Remark 13.1. Let $\mathcal{H}$ be a Hilbert space over $\mathbb{R}$ or $\mathbb{C}$ and let $\{e_\alpha\}_{\alpha \in I}$ be an orthonormal basis for $\mathcal{H}$. Suppose that for each $\alpha \in I$ there is a scalar λ_α such that $Te_\alpha = \lambda_\alpha e_\alpha$. In addition, suppose that $\varphi \in \mathcal{H}$ satisfies

$$T\varphi = \lambda\varphi$$

for some scalar $\lambda \neq \lambda_\alpha$, for all $\alpha \in I$. Then, we must have $\varphi = 0$. To see this, we fix an index $\alpha \in I$ and write

$$\begin{aligned} \lambda \langle \varphi, e_\alpha \rangle &= \langle \lambda \varphi, e_\alpha \rangle = \langle T\varphi, e_\alpha \rangle \\ &= \langle \varphi, Te_\alpha \rangle \\ &= \lambda_\alpha \langle \varphi, e_\alpha \rangle. \end{aligned}$$

Here, we have used that any eigenvalue of T must be real. Since $\lambda \neq \lambda_\alpha$, it follows that $\varphi \perp e_\alpha$. As α was arbitrary, we see that $\varphi = 0$.

Remark 13.2. An operator $T : \mathcal{H} \rightarrow \mathcal{H}$ is said to be *positive* if $\langle Tf, f \rangle > 0$ for all $f \in \mathcal{H}$ with $f \neq 0$. If T is a positive operator, then all of its eigenvalues are necessarily positive. Indeed, if $Tf = \lambda f$ for some $f \neq 0$, we have

$$0 < \langle Tf, f \rangle = \langle \lambda f, f \rangle = \lambda \|f\|^2.$$

Hence, $\lambda > 0$.

Proposition 13.3. *Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be compact, self-adjoint, and linear.*

- (1) *If λ is a non-zero eigenvalue of T , then the kernel of $T - \lambda I$ has finite dimension.*
- (2) *Let (λ_n) be an injective sequence of eigenvalues of T . Then, $\lambda_n \rightarrow 0$ as $n \rightarrow \infty$.*
- (3) *There are at most countably many eigenvalues of T .*

Proof.

- (1) Let $\lambda \neq 0$ be an eigenvalue of T and assume that $\ker(T - \lambda I)$ has infinite dimension. Then, there is a countable orthonormal family $\{\varphi_n\}_{n \in \mathbb{N}}$ in $\mathcal{H}$ with $T\varphi_n = \lambda\varphi_n$ for each $n \in \mathbb{N}$.⁷ As an orthonormal family, the φ_n are uniformly bounded. Since T is compact, after passing to a subsequence if necessarily, we may assume that $T\varphi_n$ converges. Especially, $\{T\varphi_n\}_{n \in \mathbb{N}}$ is Cauchy. On the other hand, for each $n \neq m$ there holds:

$$\|T\varphi_n - T\varphi_m\|^2 = \|\lambda\varphi_n - \lambda\varphi_m\|^2 = \lambda^2 \|\varphi_n - \varphi_m\|^2.$$

Since $\varphi_n \perp \varphi_m$, this last equation is equivalent to

$$\begin{aligned} \|T\varphi_n - T\varphi_m\|^2 &= \lambda^2 \langle \varphi_n - \varphi_m, \varphi_n - \varphi_m \rangle \\ &= \lambda^2 (\|\varphi_n\|^2 + \|\varphi_m\|^2) \\ &= 2\lambda^2 > 0. \end{aligned}$$

Hence, $\{T\varphi_n\}_{n \in \mathbb{N}}$ cannot be Cauchy in $\mathcal{H}$. This contradiction shows that the kernel of $T - \lambda I$ has finite dimension.

- (2) Let $\varepsilon > 0$ be given; we claim that there are at most finitely many eigenvalues λ of T with $|\lambda| \geq \varepsilon$. Clearly, this will imply (2). Arguing by contradiction, suppose that we can pick up a sequence $(\lambda_n)_{n \in \mathbb{N}}$ of distinct eigenvalues (with associated eigenvectors $\{\varphi_n\}_{n \in \mathbb{N}}$) such that $|\lambda_n| \geq \varepsilon$ for each $n \in \mathbb{N}$. By a previous lemma, these φ_n will be orthogonal. Thus, after a rescaling, we might as well assume they are orthonormal. In particular, $\{\varphi_n\}_{n \in \mathbb{N}}$ is a bounded sequence in $\mathcal{H}$. By compactness, we may assume (passing to a subsequence

⁷Clearly, $T - \lambda I$ is a continuous map $\mathcal{H} \rightarrow \mathcal{H}$. As such, its kernel is a closed subspace of $\mathcal{H}$ and hence must inherit a Hilbert space structure.

if necessary) that $T\varphi_n$ is Cauchy in $\mathcal{H}$. On the other hand, for all $n \neq m$ we find that:

$$\|T\varphi_n - T\varphi_m\|^2 = \|\lambda_n\varphi_n - \lambda_m\varphi_m\|^2 = \lambda_n^2 + \lambda_m^2 \geq 2\varepsilon.$$

Therefore, $\{T\varphi_n\}_{n \in \mathbb{N}}$ cannot be Cauchy in $\mathcal{H}$.

- (3) The third point follows readily from the second. If Λ denotes the set of all non-zero eigenvalues of T , then

$$\Lambda := \bigcup_{n \in \mathbb{N}} \left\{ \lambda \in \mathbb{R} : \lambda \text{ is an eigenvalue of } T \text{ and } |\lambda| \geq \frac{1}{n} \right\}.$$

By the argument used in (2), we conclude that Λ is countable.

□

An observation regarding the possible “size” of an eigenvalue is in order. Let $T : \mathcal{H} \rightarrow \mathcal{H}$ be a bounded operator (not necessarily linear or compact) and assume that $\lambda \in \mathbb{C}$ is an eigenvalue of T . Then, there is a non-zero vector $f \in \mathcal{H}$ with the property that

$$Tf = \lambda f.$$

On the other hand, $|\lambda| \|f\| = \|Tf\| \leq \|T\| \|f\|$. This means that $|\lambda| \leq \|T\|$. We now prove the spectral theorem for compact self-adjoint linear operators.

Theorem 13.4. *Let $\mathcal{H}$ be a Hilbert space over $\mathbb{R}$ or $\mathbb{C}$, and let $T : \mathcal{H} \rightarrow \mathcal{H}$ be a linear operator that is compact and self-adjoint. There exists a countable family $\{\varphi_n\}_n$ of eigenvectors, with associated eigenvalues $\{\lambda_n\}_n$ in $\mathbb{R}$, that form an orthonormal basis for $(\ker T)^\perp$. Moreover, the corresponding eigenvalues can be placed in decreasing order:*

$$|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n| \geq \cdots > 0.$$

In fact, we can define the λ_n by the equation

$$\lambda_n := \sup_{\substack{u \in H_{n-1} \\ \|u\|=1}} |\langle Tu, u \rangle| = \sup_{\substack{u \in H_{n-1} \\ \|u\|=1}} \|Tu\|$$

where H_{n-1} is defined to be the orthogonal complement of $\text{span}\{\varphi_1, \dots, \varphi_{n-1}\}$ in $\mathcal{H}$.

Proof. If $T \equiv 0$ on all of $\mathcal{H}$, then $\ker T = \mathcal{H}$ whence $\ker T^\perp = \{0\}$, at which point the claim is vacuous. Thus, we may assume that $T \not\equiv 0$. In other terms, $\|T\| > 0$ on $\mathcal{H}$ and we can find a sequence $\{v_j\}_{j \in \mathbb{N}}$ of unit vectors such that

$$|\langle Tv_j, v_j \rangle| \rightarrow \|T\| \quad \text{as } j \rightarrow \infty.$$

After passing to a subsequence if necessary, we may assume that

$$Tv_j \rightarrow w \quad \text{and} \quad \langle Tv_j, v_j \rangle \rightarrow \lambda_1;$$

where $w \in \mathcal{H}$ and $\lambda_1 \in \mathbb{C}$ is such that $|\lambda_1| = \|T\|$. Next, let us define $\varphi_1 := \lambda_1^{-1}w$. We claim that $T\varphi_1 = \lambda_1\varphi_1$. To see that this is so, write

$$\begin{aligned} \|Tv_j - \lambda_1 v_j\|^2 &= \langle Tv_j - \lambda_1 v_j, Tv_j - \lambda_1 v_j \rangle \\ &= \langle Tv_j, Tv_j - \lambda_1 v_j \rangle - \lambda_1 \langle v_j, Tv_j - \lambda_1 v_j \rangle \\ &= \|Tv_j\|^2 - \overline{\lambda_1} \langle Tv_j, v_j \rangle - \lambda_1 \langle v_j, Tv_j \rangle + |\lambda_1|^2 \|v_j\|^2 \\ &= \|Tv_j\|^2 - \overline{\lambda_1} \langle Tv_j, v_j \rangle - \lambda_1 \overline{\langle Tv_j, v_j \rangle} + |\lambda_1|^2 \|v_j\|^2 \\ &\leq |\lambda_1|^2 - \overline{\lambda_1} \langle Tv_j, v_j \rangle - \lambda_1 \overline{\langle Tv_j, v_j \rangle} + |\lambda_1|^2. \end{aligned}$$

However, we know that $\langle Tv_j, v_j \rangle \rightarrow \lambda_1$, whence $\overline{\langle Tv_j, v_j \rangle} \rightarrow \overline{\lambda_1}$. Thus, passing to the limit in the above yields

$$\|Tv_j - \lambda_1 v_j\|^2 \rightarrow |\lambda_1|^2 - \overline{\lambda_1}\lambda_1 - \lambda_1\lambda_1 + |\lambda_1|^2 = 0.$$

Since $Tv_j \rightarrow w$ in $\mathcal{H}$, the above tells us that, in fact, $\lambda_1 v_j \rightarrow w$ as $j \rightarrow \infty$. In other words, we have $v_j \rightarrow \lambda_1^{-1}w = \varphi_1$ as $j \rightarrow \infty$. Then, using the continuity of T , it follows that

$$T\varphi_1 = \lim_{j \rightarrow \infty} Tv_j = \lim_{j \rightarrow \infty} \lambda_1 v_j = w = \lambda_1 \varphi_1.$$

Since T is self-adjoint, a previous lemma shows that $\lambda_1 \in \mathbb{R}$. Note also that

$$\|\varphi_1\| = \lim_{j \rightarrow \infty} \|v_j\| = 1.$$

Consider now the subspace

$$H_1 := \{u \in \mathcal{H} : \langle u, \varphi_1 \rangle = 0\}.$$

By continuity of the inner product, this is easily seen to be a closed subspace of $\mathcal{H}$. In particular, H_1 is a Hilbert space over $\mathbb{R}$ or $\mathbb{C}$. Furthermore, T restricts to a linear map $H_1 \rightarrow H_1$. To see this, suppose that $u \in H_1$ and write

$$\langle Tu, \varphi_1 \rangle = \langle u, T\varphi_1 \rangle = \langle u, \lambda_1 \varphi_1 \rangle = \lambda_1 \langle u, \varphi_1 \rangle = 0.$$

If T is not the zero operator when restricted to H_1 , then we can repeat the procedure above to find a unit vector $\varphi_2 \in H_1$ such that

$$T\varphi_2 = \lambda_2 \varphi_2,$$

where $\lambda_2 \in \mathbb{R}$ satisfies $|\lambda_2| = \|T|_{H_1}\| > 0$. Clearly, we shall also have $|\lambda_2| \leq |\lambda_1|$ and $\varphi_2 \perp \varphi_1$. Proceeding by induction, we can construct a (possibly finite) orthonormal sequence $\{\varphi_n\}_n$ with associated eigenvalues $\{\lambda_n\}_n$ (as above) in $\mathbb{R}$. This sequence can be finite only if at some step $n \in \mathbb{N}$ we have

$$T \equiv 0 \quad \text{on} \quad H_n \quad \text{or} \quad H_n = \{0\}.$$

If $H_n = \{0\}$, then we have $\mathcal{H} = \text{span}\{\varphi_1, \dots, \varphi_n\}$. If instead $T \equiv 0$ on H_n , then we see that⁸ $H_n = \ker T$, whence we find⁹ $(\ker T)^\perp = \text{span}\{\varphi_1, \dots, \varphi_n\}$.

The proof will be complete if we can show that $\{\varphi_n\}$ forms an orthonormal basis of $(\ker T)^\perp$ in the case where the sequence $\{\varphi_n\}$ is infinite. We shall show that $\ker T = \text{span}\{\varphi_n : n \in \mathbb{N}\}^\perp$; at which point it will follow that

$$(\ker T)^\perp = \overline{\text{span}\{\varphi_n : n \in \mathbb{N}\}}.$$

Let now $u \in \ker T$ be given. Then, for every φ_j we have

$$0 = \langle Tu, \varphi_j \rangle = \langle u, \lambda_j \varphi_j \rangle = \lambda_j \langle u, \varphi_j \rangle.$$

Hence, $u \in \{\varphi_1, \dots, \varphi_n\}^\perp = \text{span}\{\varphi_n : n \in \mathbb{N}\}^\perp$. Conversely, let us fix a vector u belonging to

$$\text{span}\{\varphi_n : n \in \mathbb{N}\}^\perp.$$

Then, $\langle u, \varphi_j \rangle = 0$ for all $j \in \mathbb{N}$. In particular, $u \in H_n$ for every $n \in \mathbb{N}$. Since we have infinitely many orthonormal eigenvectors, we must have infinitely many distinct non-zero eigenvalues. Thus, we can find a subsequence (λ_{n_k}) that is injective. By previous results, we will have $\lambda_{n_k} \rightarrow 0$ as $k \rightarrow \infty$. Using that $u \in H_{n_k}$ for every k , we find that

$$\|Tu\| \leq |\lambda_{n_k}| \|u\| \rightarrow 0, \quad \text{as } k \rightarrow \infty.$$

With this, the proof is complete. $\square$

In the case where the operator T is assumed to be positive, we have a slightly stronger result.

Corollary 13.5. *Let $\mathcal{H}$ be a Hilbert space over $\mathbb{R}$ or $\mathbb{C}$ and let $T : \mathcal{H} \rightarrow \mathcal{H}$ be compact, self-adjoint, positive, and linear. There is a countable orthonormal basis for $\mathcal{H}$ consisting of eigenvectors of T .*

Proof. Since a positive operator is injective, this follows from the previous result. $\square$

14. THE EIGENVALUE PROBLEM

In this section, we will consider the *eigenvalue problem* for the Laplace operator Δ . That is, we will consider solutions to the following PDE:

$$-\Delta u \equiv \lambda u, \quad \text{in } \Omega \tag{14.1}$$

⁸Since T vanishes on H_n , we clearly have $H_n \subseteq \ker T$. Conversely, if $Tu = 0$ with $u \in \mathcal{H}$, then one has $0 = \langle Tu, \varphi_j \rangle = \langle u, T\varphi_j \rangle = \lambda_j \langle u, \varphi_j \rangle$ for all indices $j = 1, \dots, n$. Of course, this implies that $u \in H_n$.

⁹To see this, note that $H_m = \{\varphi_1, \dots, \varphi_n\}^\perp = \text{span}\{\varphi_1, \dots, \varphi_n\}^\perp$.

where $\lambda \in \mathbb{R}$ is fixed. First, however, let us put together some machinery. Let $\Omega \subset \mathbb{R}^n$ be bounded domain and let V denote either $H_0^1(\Omega)$ or $H^1(\Omega)$. As before, we define a bilinear form $V \times V \rightarrow \mathbb{R}$ via the rule

$$a : V \times V \rightarrow \mathbb{R}, \quad (u, v) \mapsto \int_{\Omega} \nabla u \cdot \nabla v.$$

Then, a weak formulation of the eigenvalue problem would involve finding $u \in V$ such that

$$a(u, v) = \lambda \langle u, v \rangle, \quad \forall v \in C_c^\infty(\Omega)$$

where $\langle \cdot, \cdot \rangle$ is the inner product on $L^2(\Omega)$. Let us now recall some basic facts:

- $a : V \times V \rightarrow \mathbb{R}$ is both symmetric and continuous in the product topology;
- There exists $t_0 \in \mathbb{R}$ with the property that the map

$$V \times V \ni (u, v) \mapsto a(u, v) + t \langle u, v \rangle$$

defines an inner product on V for all $t > t_0$. When imposing Dirichlet boundary conditions, we can take $t_0 < 0$. If we instead want Neumann boundary conditions, we can take $t_0 := 0$. Note that the map (u, v) on $V \times V$ will induce a norm equivalent to that of V .

- In fact, by more advanced results, if $a(u, v) = \lambda \langle u, v \rangle$ for all $v \in C_c^\infty(\Omega)$ then we necessarily have $u \in C^\omega(\Omega)$. That is, u will be analytic in Ω .
- If the boundary of Ω is smooth enough, then the boundary condition imposed shall be satisfied in the classical sense.

Now we return to the problem at hand. To make use of powerful results from functional analysis, we will transform the eigenvalue problem into a problem involving operators on Hilbert spaces. To this end, we define an operator

$$\tilde{A} : V \rightarrow V^*$$

by the duality pairing $\langle Au, v \rangle = a(u, v)$. Here, V^* denotes the topological dual of V . Since a is bounded, so will $\tilde{A}$. In fact, $\tilde{A}$ is symmetric in the sense that

$$\langle Au, v \rangle = a(u, v) = a(v, u) = \langle Av, u \rangle \stackrel{\text{def}}{=} \langle u, Av \rangle.$$

Let now $t > t_0$ and consider the injective linear map

$$J : L^2(\Omega) \rightarrow V^*$$

where Jf is given by $v \mapsto \int_{\Omega} f v$. Since J is injective, we can identify $L^2(\Omega)$ with a subspace of V^* . Next, we note that if $u \in C_c^\infty(\Omega)$ then

$$\tilde{A}u(v) = \int_{\Omega} \nabla u \cdot \nabla v = \int_{\Omega} v(-\Delta u), \quad \forall v \in V.$$

Thus, $\tilde{A}u$ indeed belongs to $\text{Im}(J)$ which we identify with a subspace of $L^2(\Omega)$ through J^{-1} . Define

$$D_A := \left\{ u \in V : \tilde{A}u \in \text{Im}(J) \right\} \neq \emptyset.$$

In this sense, we see that $\tilde{A}$ restricts to a well defined map

$$A : D_A \rightarrow \text{Im}(J) \rightarrow L^2(\Omega), \quad A := J^{-1} \circ \tilde{A}$$

Here, we identify $\text{Im}(J)$ with a subspace of $L^2(\Omega)$ through J^{-1} . Thus, $(A + tI)$ is a linear map $D_A \rightarrow L^2(\Omega)$. We claim this is an injection. If $(A + tI)u = Au + tu$ vanishes for $u \in D_A$, then applying J to both sides gives

$$\int_{\Omega} \nabla u \cdot \nabla v + t \int_{\Omega} uv = 0$$

for all $v \in V$. For our t , this is an inner product on V and we thus conclude that $u = 0$ almost everywhere. Hence, $(A + tI)$ is injective. Next, we show that this operator is surjective. Let $f \in L^2(\Omega)$ be given and recall that (since $t > t_0$) we can find a function $u \in V$ with

$$\int_{\Omega} \nabla u \cdot \nabla v + t \int_{\Omega} uv = \int_{\Omega} fv$$

for all $v \in V$. Hence, we have found $u \in V$ such that $\tilde{A}u + tJu \in \text{Im}(J)$. Since $tJu = J(tu)$ also belongs to the image of J , we find that $\tilde{A}u \in \text{Im}(J)$. Hence, $u \in D_A$. With this, we can safely give the following definition.

Definition 9. With $t > t_0$, we define the *resolvent* by the formula

$$R_t : L^2(\Omega) \rightarrow D_A \subseteq V \hookrightarrow L^2(\Omega), \quad R_t := (A + tI)^{-1}.$$

Heuristically, R_t looks at $f \in L^2(\Omega)$ and considers the functional $J(f) \in V^*$. The resolvent R_t then asks which function $w \in D_A \subseteq V$ is such that

$$\int_{\Omega} \nabla w \cdot \nabla v + t \int_{\Omega} wv = \int_{\Omega} fv, \quad \forall v \in V.$$

Several observations are now in order.

(1) Note that R_t is a bounded operator $L^2(\Omega) \rightarrow D_A$. Indeed, if $f \in L^2(\Omega)$ then

$$\|R_t f\|_{L^2(\Omega)} \leq \|R_t f\|_V = \|u\|_V$$

where $u \in V$ is such that

$$\int_{\Omega} \nabla u \cdot \nabla v + t \int_{\Omega} uv = \int_{\Omega} fv$$

for all $v \in V$. Thus, by the coercive property:

$$\alpha \|u\|_V^2 \leq \int_{\Omega} |\nabla u|^2 + t \int_{\Omega} u^2 = \int_{\Omega} f u$$

for a suitable $\alpha > 0$ independent of u . Thus, if $u \neq 0$:

$$\|u\|_V \leq \frac{1}{\alpha \|u\|_V} \int_{\Omega} f u \leq \frac{1}{\alpha} \|f\|_{L^2(\Omega)}.$$

Since this same inequality holds for $u = 0$, we conclude that R_t is bounded as an operator $L^2(\Omega) \rightarrow D_A$. In particular, R_t is bounded as an operator $L^2(\Omega) \rightarrow L^2(\Omega)$.

- (2) R_t is compact as an operator $L^2(\Omega) \rightarrow L^2(\Omega)$. Indeed, this follows at once from the Rellich-Kondrachov theorem.
- (3) To see that R_t is positive, let $f \in L^2(\Omega)$ be given and let $u = R_t f$, i.e.

$$\int_{\Omega} \nabla u \cdot \nabla v + t \int_{\Omega} u v = \int_{\Omega} f v, \quad \forall v \in V.$$

Then, taking $v = u$ gives

$$\int_{\Omega} |\nabla u|^2 + t \int_{\Omega} u^2 = \int_{\Omega} f u.$$

Thus,

$$\langle R_t f, f \rangle = \langle u, f \rangle = \int_{\Omega} |\nabla u|^2 + t \int_{\Omega} u^2 \geq \alpha \|u\|_V^2$$

with $\alpha > 0$ independent of u .

- (4) We claim that R_t is symmetric, i.e. that $\langle R_t f, g \rangle = \langle f, R_t g \rangle$ for any two $f, g \in L^2(\Omega)$. Denote by $\bar{f}$ and $\bar{g}$ the respective elements of D_A such that

$$\int_{\Omega} \nabla \bar{f} \cdot \nabla \varphi + t \int_{\Omega} \bar{f} \varphi = \int_{\Omega} f \varphi$$

and

$$\int_{\Omega} \nabla \bar{g} \cdot \nabla \varphi + t \int_{\Omega} \bar{g} \varphi = \int_{\Omega} g \varphi$$

for all $\varphi \in V$. Replacing φ with $\bar{g}$ in the former and putting $\varphi := \bar{f}$ in the latter shows that

$$\begin{aligned} \int_{\Omega} \nabla \bar{f} \cdot \nabla \bar{g} + t \int_{\Omega} \bar{f} \bar{g} &= \int_{\Omega} f \bar{g}, \\ \int_{\Omega} \nabla \bar{g} \cdot \nabla \bar{f} + t \int_{\Omega} \bar{g} \bar{f} &= \int_{\Omega} g \bar{f}. \end{aligned}$$

This means that

$$\int_{\Omega} f \bar{g} = \int_{\Omega} g \bar{f}.$$

Or, rather, than $\langle R_t f, g \rangle = \langle f, R_t g \rangle$.

Having established these results, we can apply the spectral theorem to R_t in a systematic way. However, we require yet *another* lemma.

Lemma 14.1. *Keeping with the context above, $\psi \in L^2(\Omega)$ is an eigenvector for R_t if and only if it is also an eigenvector for A .*

Remark 14.1. Let ψ be an eigenvector for the resolvent R_t and let λ be its corresponding eigenvalue. Since R_t is positive, we must have $\lambda > 0$. Since R_t is a map $L^2(\Omega) \rightarrow D_A$, we must have

$$R_t(\psi) = \lambda\psi \in D_A.$$

Since $\lambda \neq 0$, this means that $\psi \in D_A$. Therefore, all eigenvectors for R_t live in D_A . Hence, it makes sense to compare the eigenvectors of both R_t and A .

Proof of Lemma. Let $\psi \neq 0$ be an eigenvector of R_t with eigenvalue $\lambda > 0$. Then, we have

$$(A + tI)^{-1}(\psi) = R_t(\psi) = \lambda\psi.$$

Applying the operator $(A + tI)$ to both sides forces

$$\psi = (A + tI)(\lambda\psi) = \lambda A\psi + t\lambda\psi.$$

Solving for $A\psi$, we find that

$$A\psi = \left(\frac{1}{\lambda} - t\right)\psi.$$

In particular, ψ is an eigenvector of A with eigenvalue

$$\mu = \frac{1}{\lambda} - t. \tag{14.2}$$

Conversely, let $\psi \neq 0$ be an eigenvector of A with corresponding eigenvalue $\mu \in \mathbb{R}$. Adding $t\psi$ to both sides yields

$$(A + tI)(\psi) = A\psi + t\psi = (\mu + t)\psi.$$

Next, we apply R_t to both sides of the equation to get

$$\psi = (\mu + t) R_t(\psi).$$

Since $\psi \neq 0$, this shows in particular that $\mu + t \neq 0$. Thus,

$$R_t\psi = \frac{\psi}{\mu + t}$$

whence ψ is an eigenvector of R_t . □

We are now ready to prove our big result.

Theorem 14.2. *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain. For any $t > t_0$ there exists an orthonormal basis $\{\varphi_k\}_k$ of $L^2(\Omega)$ consisting entirely of eigenvectors of A . Denote by $\{\mu_k\}_k$ the corresponding eigenvalues in $\mathbb{R}$. Then, for every index k :*

- (1) $\ker(A - \mu_k I)$ has finite dimension;
- (2) $\mu_k > -t$;
- (3) $\mu_j \rightarrow \infty$ as $j \rightarrow \infty$.

Furthermore, if $\partial\Omega$ is sufficiently smooth, these eigenfunctions are analytic in Ω and smooth on $\bar{\Omega}$.¹⁰

Proof. We apply the spectral theorem to the resolvent R_t . In doing so, we obtain an orthonormal basis $\{\varphi_k\}_k$ for $(\ker R_t)^\perp = L^2(\Omega)$ with corresponding eigenvalues $\{\mu_k\}_k$ in $\mathbb{R}$. By the lemma with (14.2), we know that every φ_k will also be an eigenvector for A with corresponding eigenvalue

$$\lambda_k = \frac{1}{\mu_k} - t.$$

By the correspondence between eigenvectors of A and R_t , the spectral theorem immediately implies (1). By this same result, we know that $\mu_k \rightarrow 0$ as $k \rightarrow \infty$. Thus, the identity above forces $\lambda_k \rightarrow \infty$ as $k \rightarrow \infty$. Since every μ_k is positive, we must also have $\lambda_k > -t$ for any k . $\square$

Part 2. Problems and Solutions

In this part, we provide (with solutions) some highly non-trivial exercises. For the most part, these problems can be solved using arguments similar to those used in the lecture notes part of this text.

15. BOUNDARY VALUE PROBLEM UNIQUENESS EXERCISES

In what follows, Ω is a non-empty bounded domain in $\mathbb{R}^n$ for $n \geq 2$.

Problem 1. Let $u \in C^2(\Omega) \cap C(\bar{\Omega})$ be a solution to the boundary value problem:

$$\Delta u \equiv u^m \text{ in } \Omega \quad \text{and} \quad u \equiv 1 \text{ on } \partial\Omega$$

with $m \in \mathbb{N}$ odd. Prove that $0 \leq u \leq 1$ on $\bar{\Omega}$.

Proof. We first show that $u \geq 0$ in all of Ω , whence it will follow that $u \geq 0$ in $\bar{\Omega}$. Otherwise, there is a point in Ω where u is negative. In particular,

$$m := \min_{\bar{\Omega}} u < 0.$$

¹⁰We will not prove these regularity results.

Consider now the set $X := \{x \in \Omega : u(x) = m\}$. By hypothesis, this is a non-empty closed subset of Ω . If we can show that X is open in Ω , it will follow that $X = \Omega$. Since $u \equiv 1$ on $\partial\Omega$, this is impossible. Let $x \in X$ be given and find $\delta > 0$ such that $B(x, \delta) \Subset \Omega$ and $u < 0$ on the closure of $B(x, \delta)$. Since m is odd, we find that $\Delta u < 0$ on this ball. Especially, u is super-harmonic in $B(x, \delta)$. Since m must also be the minimum of u over $\overline{B(x, \delta)}$, the strong minimum principle forces $u \equiv m$ in $B(x, \delta)$. But this implies that $B(x, \delta) \subseteq X$ whence X is open.

We conclude that $u \geq 0$ so that $u \in \mathfrak{Sub}(\Omega) \cap C^2(\Omega) \cap C(\bar{\Omega})$. Applying the strong maximum principle to this function yields $u \leq 1$ in all of $\bar{\Omega}$. With this, the proof is complete. $\square$

Problem 2. Prove that any solution u to the PDE in the previous problem is unique.

Proof. Let u and v be two solutions to the aforementioned problem. Then, we put $w := u - v$ and calculate

$$\begin{aligned} \Delta w &= u^m - v^m = (u - v) [u^{m-1} + u^{m-2}v + \cdots + uv^{m-2} + v^{m-1}] \\ &= u(x)a(x), \end{aligned}$$

where by the previous problem we must have $a(x) \geq 0$ in $\bar{\Omega}$. Following the proof of in previous problem, one can easily deduce that $w \geq 0$ in Ω . Hence, w is sub-harmonic in Ω whence the strong maximum principle forces $w = u - v \leq 0$ on all of $\bar{\Omega}$. Applying the same arguments to $v - u$ gives the reverse inequality. $\square$

Problem 3. In the context of the problem above, suppose instead that u satisfies

$$u \equiv u - \frac{1}{u}$$

in the interior of Ω . Prove that $u \equiv 1$.

Proof. Note that for this problem to make sense, we must not have $u(x) = 0$ anywhere in Ω . By the intermediate value theorem, we must therefore have $u > 0$ in Ω . Next, we claim that $u \geq 1$ in all of Ω . Otherwise,

$$m := \min_{\bar{\Omega}} u$$

is achieved in the interior of Ω , and satisfies $0 < m < 1$. Now we consider the non-empty closed subset

$$X := \{x \in \Omega : u(x) = m\}$$

of Ω . To see that X is open, let $x \in X$ and choose $\delta > 0$ such that

$$B(x, \delta) \Subset \Omega \quad \text{and} \quad 0 < u(x) < 1 \text{ on } \overline{B(x, \delta)}.$$

Then, u must be super-harmonic on the closure of this ball. Since m will also be the minimum of u on $\overline{B(x, \delta)}$, the strong minimum principle implies that $u \equiv m$

on $\overline{B(x, \delta)}$. Hence, $B(x, \delta) \subseteq X$ whence X is open. By connectedness, we have a contradiction. We conclude that $u \geq 1$ in Ω . Thus, u is subharmonic in Ω and the maximum principle implies that $u \leq 1$ in Ω . This completes the proof. $\square$

Problem 4. Let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be entire harmonic for $n \geq 2$. At a any point $x_0 \in \mathbb{R}^n$, show that the tangent plane to the graph of u intersects the graph *at least* twice.

Proof. At a point $x_0 \in \mathbb{R}^n$, the equation for the tangent plane is given by the equation

$$\Gamma(x) = u(x_0) + \nabla u(x_0) \cdot (x - x_0).$$

Clearly, Γ is *also* entire harmonic in $\mathbb{R}^n$. Suppose by way of contradiction that Γ intersects the graph of u only at x_0 . Then, consider any large ball $B_R(x_0)$. We claim that $v := u - \Gamma$ cannot change signs on this ball. Otherwise, we could find a path in $B_R(x_0)$, not passing through x_0 , on which v takes both a negative and a positive value. Then, we could find $y \neq x_0$ at which v vanishes. Hence, $v \geq 0$ or $v \leq 0$ in all of space. Without loss of generality, let us consider the case where $v \geq 0$ in Ω . Since $v(x_0) = 0$, the strong maximum principle tells us that v vanishes in all of this ball. Hence, we have a contradiction. $\square$

Problem 5. Let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be an entire harmonic functions and suppose that there exists $C > 0$ and $m \in \mathbb{N}$ such that

$$u(x) \geq -C(1 + |x|)^m$$

for all $x \in \mathbb{R}^n$. Prove that u is a polynomial of degree no large than m .

Proof. Let us fix a multiindex α with $|\alpha| > m$. Consider a point $x_0 \in \mathbb{R}^n$ and let $R \gg 1$ be large. After a translation, we may assume that $u(x_0) = 0$. Then,

$$0 = \oint_{B(x_0, R)} u = \oint_{B(x_0, R)} u^+ - \oint_{B(x_0, R)} u^-.$$

Our derivative estimates show that for a suitable constant $C > 0$ we have

$$|\partial^\alpha u(x_0)| \leq \frac{C \|u\|_{L^1(x_0, R)}}{R^{n+|\alpha|}}.$$

On the other hand,

$$\begin{aligned} \int_{B(x_0, R)} |u| &= 2 \int_{B(x_0, R)} u^- \leq C \int_{B(x_0, R)} (1 + |y|)^m \\ &\leq C \int_{B(x_0, R)} (1 + |x_0 - y|)^m dy \\ &\leq C \int_0^R B(x_0, r) (1 + r)^m \\ &= C(1 + R^m)R^n. \end{aligned}$$

Putting these pieces together, we find that

$$|\partial^\alpha u(x_0)| \leq \frac{C \|u\|_{L^1(x_0, R)}}{R^{n+|\alpha|}} \leq C \frac{(1 + R^m) R^n}{R^{n+|\alpha|}}$$

which tends to zero as $R \rightarrow \infty$. This completes the proof. $\square$

Problem 6. Let $\Omega \subseteq \mathbb{R}^n$ be open and let $u \in C^1(\Omega)$ be such that, for every $x \in \Omega$, there is $r^* > 0$ such that

$$\int_{\partial B(x, r)} \partial_\nu u \, dS = 0, \quad \forall r \in (0, r^*).$$

Show that $u \in \mathfrak{H}(\Omega)$.

Proof. It is enough to prove that u satisfies the local mean value property in Ω . To this end, let $x \in \Omega$ and fix r^* as given. Without harm, we assume that $B(x, r^*) \Subset \Omega$. Define a new function

$$\eta(r) := \oint_{\partial B(x, r)} u \, dS$$

on the interval $(0, r)$. Clearly, since u is continuous on Ω , we can extend η to all of $[0, r^*]$ by taking $\eta(0) := u(x)$. Note that for all $0 < r < r^*$ one has

$$\begin{aligned} \eta'(r) &= \frac{d}{dr} \frac{1}{n\omega_n} \int_{\partial B(0, 1)} u(x + rz) \, dS(z) \\ &= \frac{1}{n\omega_n} \int_{\partial B(0, 1)} \nabla u(x + rz) \cdot z \, dS(z) \\ &= \oint_{\partial B(x, r)} \nabla u(y) \cdot \frac{y - x}{r} \, dS(y) \\ &= \int_{\partial B(x, r)} \partial_\nu u \, dS = 0. \end{aligned}$$

Hence, η is constant on $[0, r]$ with $\eta(r) = u(x)$ for each r . This means that u satisfies the local mean value property inside Ω . $\square$

16. POINCARÉ'S SWEEPING METHOD

The goal of this section is to establish an alternative to Perron's method, due to Poincaré. The setting is as follows: let $\Omega \subset \mathbb{R}^N$ be a bounded domain and let $u_0, g \in C(\bar{\Omega})$ with $u_0 \equiv g$ on $\partial\Omega$. We make the additional assumption that $u \in \mathfrak{Sub}(\Omega)$. Recall that a barrier at $z \in \partial\Omega$ is a function $\varphi \in C(\bar{\Omega}) \cap \mathfrak{Sub}(\Omega)$ such that $\varphi(z) = 0$ and $\varphi(x) < 0$ for $x \in \bar{\Omega} \setminus \{z\}$.

Problem 1. Let (X, d) be a separable metric space and let $U \subseteq X$ be an open set. Show that there exists a countable family $\{B_k\}_k$ of open balls such that $U = \bigcup_k B_k$.

Proof. Let $\Sigma = \{s_n\}_n$ be a countable dense subset of U . Given a point $x \in U$, let $\varepsilon_x \in \mathbb{Q}^+$ be such that $B(x, 2\varepsilon_x) \subseteq U$. Next, find some $s_x \in \Sigma$ with the property that $d(x, s_x) < \varepsilon_x$. By the triangle inequality, we find that

$$x \in B(s_x, \varepsilon_x) \subseteq B(x, 2\varepsilon_x) \subseteq U.$$

Thus, $U = \bigcup_{x \in U} B(s_x, \varepsilon_x)$. Since $\Sigma \times \mathbb{Q}^+$ is countable, there are only countably many terms in this union. $\square$

In particular, Ω can be written as the countable union of open balls $B_1, \dots, B_k, \dots$ where this sequence may, or may not, be finite. In particular, U can be written as the union of elements in the following sequence of balls:

$$B_1, B_2, B_1, B_2, B_3, B_1, \dots,$$

so that each B_k in our original sequence of balls is occurring infinitely many times. Let us reuse the notation B_k to denote the k^{th} sequence in this *new* sequence of balls.

Problem 2. We define functions $u_1, \dots, u_k, \dots$ in $C(\bar{\Omega})$ according to the following recursive rule:

Given $k \in \mathbb{N}$, let u_k be the solution to the Dirichlet problem

$$\begin{cases} \Delta u_k \equiv 0 & \text{in } B_k, \\ u_k \equiv u_{k-1} & \text{on } \partial B_k. \end{cases}$$

Then, extend u_k to a function $C(\bar{\Omega})$ by taking u_k to be u_{k-1} on $\bar{\Omega} \setminus B_k$.

In doing so, we obtain a function $u \in C^2(B_k) \cap C(\bar{\Omega})$.

Prove that u_k converges locally uniformly on Ω to some smooth harmonic function $u : \Omega \rightarrow \mathbb{R}$.

Proof. Since being subharmonic in Ω is a local property, it is clear by induction that each u_k will be a subharmonic function on Ω . In fact, by applying the comparison principle to the subharmonic function on $\bar{B}_k$ given by $u_{k-1} - u_k$ (which vanishes on the boundary of B_k), it is easily seen that

$$u_{k-1} \leq u_k \quad \forall k \in \mathbb{N}$$

in all of $\bar{\Omega}$. Thus, our recursive procedure yields an increasing sequence of subharmonic functions defined on $\bar{\Omega}$. Furthermore, by the strong maximum principle for subharmonic functions, we actually have that $u_k \leq \max_{\partial\Omega} g$ on $\bar{\Omega}$, for all $k \in \mathbb{N}$. Consequently, the expression

$$u(x) := \lim_{k \rightarrow \infty} u_k(x)$$

gives us a well defined function $u : \bar{\Omega} \rightarrow \mathbb{R}$. We will prove that $u \in \mathfrak{H}(\Omega)$ and that u_k converges locally uniformly to u in Ω . To achieve this, let us pick a ball B_ℓ in our

original *non-repeating* sequence of balls. Since the collection of all such balls cover Ω , it is enough to show that u_k converges locally uniformly to u on B_ℓ . Since the ball B_ℓ occurs infinitely many times in our new sequence, we can pick up a subsequence $\{u_{k_m}\}_{m \in \mathbb{N}}$ such that

- each u_{k_m} is harmonic in B_ℓ ,
- every u_{k_m} is continuous on $\bar{\Omega}$ and subharmonic in Ω .

By Harnack's principle, u_{k_m} must converge locally uniformly on B_ℓ to a function that is harmonic in B_ℓ . However, since u_{k_m} converges pointwise to u on all of $\bar{\Omega}$, the only possibility is that u_{k_m} converges locally uniformly to u on B_ℓ . Of course, this also tells us that u is harmonic in B_ℓ , and hence on Ω . All that remains is to check that the *entire sequence* converges locally uniformly to u on B_ℓ . However, this is really just a consequence of the pointwise monotonicity of the sequence. Indeed, if K is a compact subset of B_ℓ and $\varepsilon > 0$ is given, we can find $M \in \mathbb{N}$ such that

$$\sup_{x \in K} |u(x) - u_{k_m}(x)| < \varepsilon$$

for all $m \geq M$. Then, if $k \geq k_M$ we find that

$$|u(x) - u_k(x)| = u(x) - u_k(x) \leq u(x) - u_{k_M}(x) \leq \varepsilon$$

for every $x \in K$. Thus, the claim follows. $\square$

Note that u will be identically equal to g on $\partial\Omega$, as u_k will be equal to g on all of $\partial\Omega$ at every stage $k \in \mathbb{N}$.

Problem 3. Suppose that there exists $v \in C^2(\Omega) \cap C(\bar{\Omega})$ solving the boundary value problem:

$$\begin{cases} \Delta v \equiv 0 & \text{in } \Omega, \\ v \equiv g & \text{on } \partial\Omega. \end{cases}$$

Prove that we will have $u \equiv v$ on all of $\bar{\Omega}$.

Proof. The only non-trivial claim is the equality $u \equiv v$ in Ω . Fix $k \in \mathbb{N}$ and note that $u_k - v$ is continuous on $\bar{\Omega}$ and subharmonic in Ω (v is harmonic in Ω). Since $u_k - v$ vanishes on $\partial\Omega$, the strong maximum principle forces

$$u_k \leq v$$

pointwise everywhere on $\bar{\Omega}$. Passing to the limit as $k \rightarrow \infty$, we see that

$$u(x) = \lim_{k \rightarrow \infty} u_k(x) \leq v(x)$$

on all of $\bar{\Omega}$. For the reverse inequality, note that the function $v - u$ is subharmonic in all of Ω . Since $u_k \leq u$ in all of $\bar{\Omega}$ by construction, it follows that

$$\limsup_{\substack{x \in \Omega \\ x \rightarrow z}} (v - u) = g(z) - \liminf_{\substack{x \in \Omega \\ x \rightarrow z}} u \leq g(z) - \liminf_{\substack{x \in \Omega \\ x \rightarrow z}} u_k = 0$$

for all $k \in \mathbb{N}$. Hence, the strong maximum principle tells us that $v \leq u$ on all of Ω . Since $u = v$ on Ω , all of these inequalities combined gives that $u \equiv v$ on $\bar{\Omega}$. $\square$

Problem 4. Fix $z \in \partial\Omega$ and assume that there exists a barrier at z . Prove that

$$\lim_{\substack{x \in \Omega \\ x \rightarrow z}} u(x) = g(z).$$

Proof. Since $u \geq u_k$ in all of $\bar{\Omega}$, for each $k \in \mathbb{N}$, we see that

$$\liminf_{\substack{x \in \Omega \\ x \rightarrow z}} u \geq \liminf_{\substack{x \in \Omega \\ x \rightarrow z}} u_k = g(z).$$

On the other hand, let us fix $\varepsilon > 0$ and $k \in \mathbb{N}$. Since z is a regular point, we can find $w \in \mathcal{S}_{-g}$ such that

$$w(z) \geq -g(z) - \varepsilon.$$

Then, for every index $k \in \mathbb{N}$ the sum $u_k + w$ will be harmonic in Ω , with $u_k + w \leq 0$ on $\partial\Omega$. By the maximum principle, we must have $u_k \leq -w$ on all of $\bar{\Omega}$. Letting $k \rightarrow \infty$, we get that $u \leq -w$ on $\bar{\Omega}$. Consequently,

$$\limsup_{\substack{x \in \Omega \\ x \rightarrow z}} u \leq -w(z) \leq g(z) + \varepsilon.$$

Sending $\varepsilon \rightarrow 0$ yields the desired result. $\square$

Problem 5. Let $\{v_j\}_{j \in \mathbb{N}}$ be a sequence of harmonic functions in Ω , all continuous up to the boundary of Ω , and assume that this sequence converges uniformly on $\partial\Omega$ to some continuous map $g : \partial\Omega \rightarrow \mathbb{R}$. Show that there exists a harmonic function $u \in C(\bar{\Omega})$ which satisfies $u \equiv g$ on $\partial\Omega$.

Proof. We will first show that $\{v_j\}$ converges uniformly on Ω . By Harnack's theorems, this will ensure the existence of a continuous function $u : \bar{\Omega} \rightarrow \mathbb{R}$ that is harmonic in the interior of Ω . Given any indices j and ℓ in $\mathbb{N}$, the strong maximum principle tells us that

$$v_j(x) - v_\ell(x) \leq \sup_{\partial\Omega} |v_j - v_\ell|$$

for all $x \in \bar{\Omega}$. Similarly, the strong minimum principle forces

$$v_j(x) - v_\ell(x) \geq -\sup_{\partial\Omega} |v_j - v_\ell|, \quad \forall x \in \bar{\Omega}.$$

Hence,

$$\sup_{\bar{\Omega}} |v_j - v_\ell| \leq \sup_{\partial\Omega} |v_j - v_\ell|.$$

Thus, v_j converges uniformly on $\bar{\Omega}$ to some continuous map $u : \bar{\Omega} \rightarrow \mathbb{R}$ that is harmonic in Ω . To see that $u \equiv g$ on $\partial\Omega$, fix a point $z \in \partial\Omega$ and note that

$$u(z) = \lim_{j \rightarrow \infty} v_j(z) = g(z)$$

where we have used here that $v_j \rightarrow g$ uniformly on $\partial\Omega$. $\square$

17. SOBOLEV SPACES ON INTERVALS

In the following problems, we consider the Sobolev space $W^{1,p}(I)$, where I is the unit interval $(0,1) \subset \mathbb{R}$ and $1 \leq p < \infty$. However, we will build up this space in a more abstract setting. Instead of directly working with $W^{1,p}(I)$, we will instead work with $H^{1,p}(I)$, where this space is defined to be the *completion* of $C^1(\bar{I})$ with respect to the norm

$$\|u\|_{1,p} := \left(\int_I |u|^p + \int_I |u'|^p \right)^{1/p}.$$

Problem 1. Prove that there exists an embedding $H^{1,p}(I) \hookrightarrow L^p(I)$.

Proof. Let us fix an element u of $H^{1,p}(I)$ and choose a sequence $\{\phi_k\}_{k \in \mathbb{N}}$ in $C^1(\bar{I})$ representing u . Then, ϕ_k converges to u relative to the $H^{1,p}(I)$ norm. In particular, ϕ_k is Cauchy in $H^{1,p}(I)$. Clearly, this means that $\{\phi_k\}$ will be Cauchy in $L^p(I)$. Thus, we can find $\phi \in L^p(I)$ such that $\phi_k \rightarrow \phi$ in $L^p(I)$. Similarly, ϕ'_k converges in $L^p(I)$ to some $\phi' \in L^p(I)$. This induces two “inclusion type” mappings

$$J_0 : H^{1,p}(I) \rightarrow L^p(I), \quad u \mapsto \phi, \tag{17.1}$$

$$J_1 : H^{1,p}(I) \rightarrow L^p(I), \quad u \mapsto \phi'. \tag{17.2}$$

Let us check that J_0 and J_1 are well defined. Suppose that $\{\phi_k\}$ and $\{\varphi_k\}$ are two sequences in $C^1(\bar{I})$ representing the same element in $H^{1,p}(I)$. Thus, both converge to u in $H^{1,p}(I)$. Of course, this means that

$$\phi_k - \varphi_k$$

converges to 0 in $H^{1,p}(I)$ as $k \rightarrow \infty$. In particular, $\phi_k - \varphi_k$ and $\phi'_k - \varphi'_k$ converge to 0 in $L^p(I)$ as $k \rightarrow \infty$. It follows that $\{\phi_k\}$ and $\{\varphi_k\}$ have the same limit in $L^p(I)$. Thus, J_0 is well defined. In a similar vein, we see that J_1 is well defined. Next, we consider the coordinate map

$$J : H^{1,p}(I) \rightarrow L^p(I) \times L^p(I), \quad u \mapsto (J_0(u), J_1(u)).$$

We claim that J is injective. To see this, assume that $J(u) = J(v)$ for any two $u, v \in H^{1,p}(I)$. Let $\{\phi_k\}$ and $\{\psi_k\}$ be sequences in $C^1(\bar{I})$ representing u and v ,

respectively. Since $J_0(u) = J_0(v)$, both ϕ_k and ψ_k have the same limit in $L^p(I)$. Similarly, so do ϕ'_k and ψ'_k . Therefore, the mixed sequence

$$\phi_1, \psi_1, \dots, \phi_k, \psi_k, \dots$$

is Cauchy in $H^{1,p}(\Omega)$ with some limit w . Then, every subsequence of this mixed sequence will also converge to w . It follows that $u = v$. Thus, J is an injective map. Using this, we will actually show that J_0 is injective. Since J is injective, it is enough to show that $J_0(u) = J_0(v)$ forces $J_1(u) = J_1(v)$.

To this end, let $J_0(u) = J_0(v)$ and let $\{\phi_k\}, \{\psi_k\}$ be sequences in $C^1(\bar{I})$ representing u and v , respectively. By hypothesis, $\phi_k - \psi_k \rightarrow 0$ in $L^p(I)$ as $k \rightarrow \infty$. Let

$$\alpha := \lim_{k \rightarrow \infty} \phi'_k \quad \text{and} \quad \beta := \lim_{k \rightarrow \infty} \psi'_k,$$

where both these limits are taken in L^p -sense. Now, let us fix $\varphi \in C_c^\infty(\bar{I})$. We calculate by way of Hölder's inequality:

$$\begin{aligned} \int_{\Omega} \alpha \varphi &= \lim_{k \rightarrow \infty} \int_{\Omega} \phi'_k \varphi = - \lim_{k \rightarrow \infty} \int_{\Omega} \phi_k \varphi' \\ &= - \lim_{k \rightarrow \infty} \int_{\Omega} \psi_k \varphi' \\ &= \lim_{k \rightarrow \infty} \int_{\Omega} \psi'_k \varphi \\ &= \int_{\Omega} \beta \varphi. \end{aligned}$$

Since φ was taken arbitrarily, we see that $\alpha = \beta$ almost everywhere on $\bar{I}$. Hence, $J_1(u) = J_1(v)$ and so $u = v$. This proves that J_0 is injective. Clearly, J_0 is linear. All that remains is to prove that J_0 is bounded. However, this is easily seen by the continuity of the norm: if $u \in H^{1,p}(I)$ and $\{\phi_k\} \subset C^1(\bar{I})$ represents u , then

$$\|J_0(u)\|_{L^p(I)} = \lim_{k \rightarrow \infty} \|\phi_k\|_{L^p(I)} \leq \lim_{k \rightarrow \infty} \|\phi_k\|_{H^{1,p}(I)} = \|u\|_{H^{1,p}(I)}.$$

Thus, $J_0 : H^{1,p}(I) \hookrightarrow L^p(I)$. □

Problem 2. Let $a, b \geq 0$ and $1 \leq p < \infty$. Prove that

$$(a + b)^p \leq 2^{p-1}(a^p + b^p).$$

Proof. Consider the function $f(x) = x^p$; by the second derivative test, this function is convex on $[0, \infty)$. By convexity, it follows that

$$(ta + (1 - t)b)^p \leq ta^p + (1 - t)b^p, \quad \forall t \in [0, 1].$$

In particular, for $t = \frac{1}{2}$ we see that

$$\frac{(a+b)^p}{2^p} \leq \frac{a^p + b^p}{2}$$

whence the claim follows. $\square$

Problem 3. Prove that

$$\|u\|_{L^\infty(I)} \leq 2^{1-1/p} \|u\|_{H^{1,p}(I)}$$

for all $u \in H^{1,p}(I)$.

Proof. By a previous problem, we can choose to view u as a function in $L^p(I)$. First, we prove the claim for $u \in C^1(\bar{I})$. Fixing $x \in \bar{I}$, we note that

$$\begin{aligned} |u(x)| &= \int_I |u(x)| \, dy \leq \int_I |u(x) - u(y)| \, dy + \int_I |u(y)| \, dy \\ &= \int_I |u| + \int_I \left| \int_y^x u' \right| \\ &\leq \int_I |u| + \int_I \int_I |u'| \\ &= \int_I |u| + \int_I |u'| \\ &\leq \|u\|_{L^p(I)} + \|u'\|_{L^p(I)}. \end{aligned}$$

Note that the last step follows from Hölder's inequality. Finally, an application of the previous problem shows that

$$|u(x)| \leq \|u\|_{L^p(I)} + \|u'\|_{L^p(I)} \leq 2^{1-1/p} \left(\|u\|_{L^p(I)}^p + \|u'\|_{L^p(I)}^p \right)^{1/p}$$

whence

$$|u(x)| \leq 2^{1-1/p} \|u\|_{H^{1,p}(I)}$$

for all $x \in \bar{I}$. Now, we extend this inequality to $u \in H^{1,p}(I) \subseteq L^p(I)$. Given such a function, we can find a sequence $\{u_k\}$ in $C^1(\bar{I})$ such that $u_k \rightarrow u$ in $H^{1,p}(I)$. In particular, $\|u_k\|_{H^{1,p}(I)} \rightarrow \|u\|_{H^{1,p}(I)}$. Since convergence in $H^{1,p}(I)$ implies convergence in $L^p(I)$, after passing to a subsequence if necessary, we can assume that $u_k \rightarrow u$ pointwise almost everywhere on $\bar{I}$. For any such x , we find that

$$|u(x)| = \lim_{k \rightarrow \infty} |u_k(x)| \leq \lim_{k \rightarrow \infty} 2^{1-1/p} \|u_k\|_{H^{1,p}(I)} = 2^{1-1/p} \|u\|_{H^{1,p}(I)}.$$

Since this holds for almost all $x \in I$, the claim follows. $\square$

Problem 4. Let $u \in H^{1,p}(I)$ be given. Show that u is continuous, up to modification on a set of Lebesgue measure zero.

Proof. Select a sequence $\{u_k\}$ in $C^1(\bar{I})$ representing u , i.e. such that $u_k \rightarrow u$ in $H^{1,p}(I)$. Again, we may assume without loss of generality that u_k converges pointwise almost everywhere to u on I . Now, $\{u_k\}$ will be Cauchy in $H^{1,p}(I)$. By the previous problem, it follows that u_k is Cauchy in $L^\infty(I) \cong L^\infty(\bar{I})$. By continuity, u_k is actually Cauchy in the uniform norm on $[0, 1]$. Thus, u_k converges uniformly to a function $w \in C^1(\bar{I})$. Since this implies pointwise convergence, we get that $u(x) = w(x)$ for almost all $x \in (0, 1)$. Equivalently, $u(x) = w(x)$ for almost all $x \in [0, 1]$. This completes the proof. $\square$

Henceforth, when we speak of functions $u \in H^{1,p}(I) \subseteq L^p(I)$, we will always consider a continuous representative, i.e. we assume $u \in C(\bar{I})$.

Problem 5. Prove that for any $u \in H^{1,p}(I)$ and any $\xi \in [0, 1]$, there holds the following:

$$\|u\|_{L^p(I)}^p \leq 2^{p-1} \|u'\|_{L^p(I)}^p + 2^{p-1} |u(\xi)|^p.$$

Proof. Let us first establish the inequality for $u \in C^1(\bar{I})$. For $x \in [0, 1]$, we have

$$\begin{aligned} |u(x)| &\leq |u(x) - u(\xi)| + |u(\xi)| \\ &= \left| \int_{\xi}^x u' \right| + |u(\xi)| \\ &\leq \int_I |u'| + |u(\xi)| \\ &\leq \|u'\|_{L^p(I)} + |u(\xi)|. \end{aligned}$$

Raising both sides to the power p and applying Problem 2, we see that

$$|u(x)|^p \leq \left(\|u'\|_{L^p(I)} + |u(\xi)| \right)^p \leq 2^{p-1} \left(\int_I |u'|^p + |u(\xi)|^p \right).$$

Integrating both sides of this equation yields the desired result. Thus, the claim is verified for $u \in C^1(\bar{I})$. Next, we let $u \in H^{1,p}(I)$ be arbitrary. However, by the last problem, we can choose a representative of u that is continuous on $[0, 1]$. Letting $\{u_k\}$ be a sequence in $C^1(\bar{I})$ converging to u in $H^{1,p}(I)$, Problem 3 shows that u_k will converge uniformly to u on $\bar{I}$. Therefore, for any $x \in [0, 1]$ we have

$$\begin{aligned} |u(x)|^p &= \lim_{k \rightarrow \infty} |u_k(x)|^p \leq \lim_{k \rightarrow \infty} 2^{p-1} \left(\int_I |u'_k|^p + |u_k(\xi)|^p \right) \\ &= 2^{p-1} \left(\int_I |u'|^p + |u(\xi)|^p \right). \end{aligned}$$

This completes the proof. $\square$

Problem 6. Show that

$$\|u\|_{L^p(I)}^p \leq 2^{p-1} \|u'\|_{L^p(I)}^p + 2^{p-1} \left| \int_I u \right|^p$$

for every $u \in H^{1,p}(I)$.

Proof. As before, we might as well assume that u is continuous on $\bar{I}$. In light of the previous problem, it is enough to show that there is $\xi \in [0, 1]$ with the property that

$$u(\xi) = \int_I u.$$

Since u is continuous on the compact set $\bar{I}$, it achieves both a global maximum and a global minimum. Let x_* and x^* in $\bar{I}$ be such that

$$u(x^*) = \max_{\bar{I}} u \quad \text{and} \quad u(x_*) = \min_{\bar{I}} u.$$

Now, consider function

$$g : [0, 1] \rightarrow \mathbb{R}, \quad g(x) = u(x) - \int_I u.$$

Clearly, this map is continuous. Furthermore, since

$$u(x_*) = \int_I u(x_*) \leq \int_I u, \tag{17.3}$$

$$u(x^*) = \int_I u(x^*) \geq \int_I u, \tag{17.4}$$

the intermediate value theorem ensures the existence of $\xi \in [0, 1]$ with $g(\xi) = 0$. That is, such that $u(\xi) = \int_I u$. $\square$

18. MINIMIZATION OF ENERGIES

In this section we practice the variational approach to determining the existence of solutions to PDE. Namely, we interpret solutions to PDE as minimizers of the appropriate energy functionals.

Problem 1. Let $\mathcal{A}$ be the set of all $C^1(I) \cap C^1(\bar{I})$ functions u satisfying $u(0) = 0$ and $u(1) = 1$, where $I = (0, 1)$. For a function $u \in \mathcal{A}$, we define its energy as follows:

$$E(u) := \int_I (1 + u'(x)^2)^{1/4} dx.$$

Calculate $\inf_{u \in \mathcal{A}} E(u)$ but show that the infimum is never attained.

Proof. Taking $u(x) := x$, it is easy to see that the infimum will be finite. As a first step, we show that $E(u) \geq 1$ for all admissible $u \in \mathcal{A}$. To see this, note that for any such u there holds:

$$E(u) \geq \int_I 1 \, dx = 1.$$

Second, we show that there exists a sequence $\{u_n\}$ in $\mathcal{A}$ such that $E(u_n) \rightarrow 1$ as $n \rightarrow \infty$. Clearly, this would imply that

$$\inf_{u \in \mathcal{A}} E(u) = 1.$$

To this end, let us define $u_n(x) := x^n$ for each $n \in \mathbb{N}$. Then, $u_n \in \mathcal{A}$ and satisfies

$$\begin{aligned} 1 \leq E(u_n) &= \int_I (1 + n^2 x^{2(n-1)})^{1/4} \, dx \\ &\leq \int_I \left(1 + \sqrt{n} x^{\frac{n-1}{2}}\right) \, dx \\ &\leq 1 + \sqrt{n} \int_0^1 x^{\frac{n-1}{2}} \, dx \\ &= 1 + \sqrt{n} \cdot \frac{2}{n+1} \xrightarrow{n \rightarrow \infty} 1. \end{aligned}$$

Thus, the infimum of $E(\cdot)$ over $\mathcal{A}$ is indeed equal to 1. On the other hand, if $E(u) = 1$ for some $u \in \mathcal{A}$, we would have

$$0 = \int_I [(1 + (u')^2)^{1/4} - 1] \, dx = 0$$

so that $1 + (u')^2 \equiv 1$ on I . Obviously, this would force u to be constant on I , which is absurd. $\square$

Problem 2. Let $\Omega \subseteq \mathbb{R}^n$ be open and let $u \in C^2(\Omega)$ be such that

$$E(u) := \int_{\Omega} (|\nabla u|^2 + u^2) < \infty.$$

Prove that the following two statements are equivalent:

- (1) $\Delta u \equiv u$ in Ω ;
- (2) for every test function $v \in C_c^\infty(\Omega)$, there holds $E(u) \leq E(u + v)$.

Proof. Suppose first that $\Delta u \equiv u$ in Ω . Given a test function $v \in C_c^\infty(\Omega)$, we compute

$$\begin{aligned} E(u+v) - E(u) &= \int_{\Omega} |\nabla u + \nabla v|^2 + (u+v)^2 - \int_{\Omega} |\nabla u|^2 - \int_{\Omega} u^2 \\ &= E(v) + 2 \int_{\Omega} \nabla u \cdot \nabla v + 2 \int_{\Omega} uv \\ &\geq 2 \int_{\Omega} \nabla u \cdot \nabla v + 2 \int_{\Omega} uv \\ &= 2 \left(\int_{\Omega} uv - \int_{\Omega} v \Delta u \right) = 0. \end{aligned}$$

Thus, we see that (2) holds. Conversely, suppose that (2) holds and fix $v \in C_c^\infty(\Omega)$. Given $\varepsilon \in \mathbb{R}$, we compute the following:

$$0 \leq E(u + \varepsilon v) - E(u) = 2\varepsilon \int_{\Omega} (\nabla u \cdot \nabla v + uv) + \varepsilon^2 E(v).$$

Integration by parts tells us that

$$0 \leq 2\varepsilon \int_{\Omega} (u - \Delta u)v + \varepsilon^2 E(v), \quad \forall \varepsilon \in \mathbb{R}.$$

Of course, it follows that

$$\varepsilon \int_{\Omega} (\Delta u - u)v \leq \frac{\varepsilon^2}{2} E(v), \quad \forall \varepsilon \in \mathbb{R}.$$

This condition forces $\int_{\Omega} (\Delta u - u)v = 0$, and since v was arbitrary, we must have $\Delta u \equiv u$ in Ω . $\square$

19. REGULARITY AND DIFFERENTIABILITY

Let $\Omega \subseteq \mathbb{R}^n$ be open and let $f \in L_{\text{loc}}^2(\Omega)$. A function $g_i \in L_{\text{loc}}^2(\Omega)$ is said to be the *strong L^2 - i^{th} derivative of f* if for every compact set $K \subset \Omega$, we can find a sequence $\{\varphi_k\}_{k \in \mathbb{N}}$ in $C^\infty(K)$ such that

$$f = \lim_{k \rightarrow \infty} \varphi_k \quad \text{and} \quad g_i = \lim_{k \rightarrow \infty} \partial_i \varphi_k,$$

with the convergence taking place in $L^2(\Omega)$. Similarly, one can define strong L^p -derivatives for all $1 \leq p < \infty$. The next few exercises deal with strong L^2 -derivatives and their relationship with weak derivatives.

Problem 1. Let $f \in L^2_{\text{loc}}(\Omega)$ and let g_i be the strong L^2 - i^{th} derivative of f in Ω . Then, for every test function $\varphi \in C_c^\infty(\Omega)$, we have the integration by parts identity:

$$\int_{\Omega} f \partial_i \varphi = - \int_{\Omega} g_i \varphi.$$

Furthermore, g_i is uniquely determined in the $L^2(\Omega)$ sense.

Proof. Let us fix a test function φ on Ω and denote by K the compact support of φ in Ω . We may pick a sequence $\{\phi_k\}_{k \in \mathbb{N}}$ in $C^\infty(K)$ such that $\phi_k \rightarrow f$ in $L^2(K)$ and $\partial_i \phi_k \rightarrow g_i$ in $L^2(K)$, as $k \rightarrow \infty$. By Hölder's inequality, it is easily seen that

$$\begin{aligned} \int_{\Omega} f \partial_i \varphi &= \int_K f \partial_i \varphi = \lim_{k \rightarrow \infty} \int_K \phi_k \partial_i \varphi \\ &= \lim_{k \rightarrow \infty} \int_{\Omega} \phi_k \partial_i \varphi. \end{aligned}$$

On the other hand,

$$\lim_{k \rightarrow \infty} \int_{\Omega} \phi_k \partial_i \varphi = - \lim_{k \rightarrow \infty} \int_K \varphi \partial_i \phi_k = - \int_K \varphi g_i$$

which is precisely equal to

$$- \int_{\Omega} \varphi g_i.$$

This establishes the integration by parts identity. For the uniqueness part, suppose that h_i is another strong L^2 - i^{th} derivative of f on Ω . Then, for every $\varphi \in C_c^\infty(\Omega)$ we have that

$$\int_{\Omega} g_i \varphi = - \int_{\Omega} f \partial_i \varphi = \int_{\Omega} h_i \varphi.$$

From this it follows that $f_i = h_i$ almost everywhere in Ω . In particular, we see that L^2 -strong derivatives are also weak derivatives. $\square$

Problem 2. Let $\Omega \subseteq \mathbb{R}^n$ be open and let $f \in L^2_{\text{loc}}(\Omega)$. Prove that f has an i^{th} weak derivative $\partial_i f$ on Ω if and only if f is strongly L^2 -differentiable in the i^{th} direction with strong derivative equal to $\partial_i f$.

Proof. One direction was proven in the previous exercise. Conversely, suppose that f has a weak derivative $\partial_i f$ in Ω . Fix a compact set $K \subset \Omega$ and let f_ε denote the mollified version of f on Ω . For all $\varepsilon > 0$ small, we will have $f_\varepsilon \in C^\infty(K)$. Furthermore, f_ε will converge to f in $L^2(K)$ as $\varepsilon \rightarrow 0$. Now, for all $\varepsilon > 0$ sufficiently

small and any $x \in K$ we can calculate:

$$\begin{aligned}\partial_i f_\varepsilon(x) &= \int_{\Omega} \partial_{x_i} \eta_\varepsilon(x-y) f(y) \, dy = - \int_{\Omega} \partial_{y_i} \eta_\varepsilon(x-y) f(y) \, dy \\ &= \int_{\Omega} \eta_\varepsilon(x-y) \partial_i f(y) \, dy\end{aligned}$$

where this last term is precisely the mollified version of $\partial_i f$. Since this will also converge to $\partial_i f$ in $L^2(K)$, the proof is complete. $\square$

Finally, we extend a regularity result for weakly harmonic functions.

Problem 3. Let $\Omega \subseteq \mathbb{R}^n$ be open and let $f \in L^1_{\text{loc}}(\Omega)$ be such that

$$\int_{\Omega} u \Delta \varphi = 0, \quad \forall \varphi \in C_c^\infty(\Omega).$$

After a modification on a set of measure zero, φ is classically harmonic in Ω .

Proof. Let $\{K_j\}_{j=1}^\infty$ be an exhaustion of Ω by compact sets. Then, every K_j is contained within the interior of K_{j+1} and $K_j \nearrow \Omega$. Fixing a compact set $K \subseteq \Omega$ with non-empty interior, we know that the mollified version u_ε of u will be smooth on K for all $\varepsilon > 0$ small. In fact, by differentiating under the integral, each u_ε will be harmonic on K and, especially, in the interior of K . By the mean value property, it is not hard to see that the u_ε are uniformly locally Cauchy in the interior of K . Hence, the u_ε converge locally uniformly to a function $w : \text{Int}(K) \rightarrow \mathbb{R}$ that is harmonic in the interior of K . Since the u_ε converge to u in $L^1(K)$, we see that u is equal almost everywhere to a harmonic function in the interior of K . Applying this argument to each term of our exhaustion completes the proof. $\square$

20. ON ELLIPTIC OPERATORS

In this section, we study an elliptic problem. Working in $\mathbb{R}^n$, for $n \geq 3$, let $(a_{ij})_{i,j=1}^n$ be a matrix with smooth entries, and suppose that $b : \mathbb{R}^n \rightarrow \mathbb{R}$ is smooth. Denote by Ω a smoothly bounded domain in $\mathbb{R}^n$. Let us define the following bilinear form:

$$a : H^1(\Omega) \times H^1(\Omega) \rightarrow \mathbb{R}, \quad (u, v) \mapsto \int_{\Omega} \left(buv + \sum_{i,j=1}^n a_{ij} \partial_i u \partial_j v \right).$$

Henceforth, we assume that the matrix (a_{ij}) is symmetric and that it satisfies the *uniform ellipticity condition*

$$\sum_{i,j=1}^n a_{ij}(x) \xi_i \xi_j \geq \lambda |\xi|^2, \quad \forall x \in \bar{\Omega}, \forall \xi \in \mathbb{R}^n \quad (20.1)$$

with $\lambda > 0$ a constant.

Problem 1. Prove that the map

$$A : H^1(\Omega) \rightarrow (H^1(\Omega))^*, \quad u \mapsto \langle Au, \cdot \rangle,$$

where for all $v \in H^1(\Omega)$:

$$\langle Au, v \rangle = a(u, v) = \int_{\Omega} \left(buv + \sum_{i,j=1}^n a_{ij} \partial_i u \partial_j v \right),$$

is a bounded operator. Here, $(H^1(\Omega))^*$ denotes the topological dual of $H^1(\Omega)$.

Proof. Let $u \in H^1(\Omega)$ be given, and note that $\langle Au, \cdot \rangle$ is a linear functional on $H^1(\Omega)$. Assume that $u \in H^1(\Omega)$ is a unit vector, and observe that for

$$M = \sup_{x \in \bar{\Omega}} \{|a_{ij}(x)|, |b(x)|\} \geq 0$$

there holds

$$\begin{aligned} |\langle Au, v \rangle| &\leq \int_{\Omega} |buv| + \sum_{i,j=1}^n \int_{\Omega} |a_{ij} \partial_i u \partial_j v| \\ &\leq M \int_{\Omega} |u| |v| + M \sum_{i,j=1}^n \int_{\Omega} |\partial_i u| |\partial_j v| \\ &\leq M \|u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + M \sum_{i,j=1}^n \|\partial_i u\|_{L^2(\Omega)} \|\partial_j v\|_{L^2(\Omega)} \\ &\leq M \|u\|_{H^1(\Omega)} + Mn^2 \|u\|_{H^1(\Omega)} \\ &= M(n^2 + 1) \|u\|_{H^1(\Omega)}. \end{aligned}$$

Thus, we see that $\|Au\|_{\text{op}} \leq M(n^2 + 1) \|u\|_{H^1(\Omega)}$ for all $u \in H^1(\Omega)$. It follows that A is a bounded operator. $\square$

Problem 2. Assuming, additionally, that $b > 0$ in all of $\bar{\Omega}$, show that there exists a constant $\gamma > 0$ such that

$$a(u, u) \geq \gamma \|u\|_{H^1(\Omega)}^2$$

for all $u \in H^1(\Omega)$.

Proof. Let $m > 0$ denote the absolute minimum that b achieves on $\bar{\Omega}$. Given a function $u \in H^1(\Omega)$ we compute

$$\begin{aligned} a(u, u) &= \int_{\Omega} bu^2 + \sum_{i,j=1}^n \int_{\Omega} a_{ij} \partial_i u \partial_j u \\ &\geq m \int_{\Omega} u^2 + \int_{\Omega} \sum_{i,j=1}^n a_{ij} \partial_i u \partial_j u \\ &\geq m \|u\|_{L^2(\Omega)}^2 + \lambda \int_{\Omega} |\nabla u|^2, \end{aligned}$$

where we have used the uniform ellipticity condition in (20.1) for this last inequality. Setting $\gamma > 0$ to be the minimum of both λ and m , we find that

$$a(u, u) \geq \gamma \left(\|u\|_{L^2(\Omega)}^2 + |\nabla u|^2 \right) = \gamma \|u\|_{H^1(\Omega)}^2.$$

□

Problem 3. Again assuming that $b > 0$ in $\bar{\Omega}$, show that the bilinear form

$$(\cdot, *) : H^1(\Omega) \times H^1(\Omega) \rightarrow \mathbb{R}, \quad (u, v) \mapsto a(u, v)$$

is an inner product on $H^1(\Omega)$ inducing an equivalent topology.

Proof. Arguing as in problem 1, it is not hard to see that there exists a constant $M > 0$ such that

$$a(u, u) \leq M \|u\|_{H^1(\Omega)}^2$$

for all $u \in H^1(\Omega)$. Since (a_{ij}) is a symmetric matrix, it is clear that $(u, v) = (v, u)$ for all $u, v \in H^1(\Omega)$. By the previous problem, there is $\gamma > 0$ such that

$$\gamma \|u\|_{H^1(\Omega)}^2 \leq (u, u) \leq M \|u\|_{H^1(\Omega)}^2, \quad \forall u \in H^1(\Omega).$$

Thus, $(u, u) = 0$ if and only if $u = 0$ in $H^1(\Omega)$. This means that $(\cdot, *)$ is an inner product on $H^1(\Omega)$. By the above, it induces a norm equivalent to $\|\cdot\|_{H^1(\Omega)}$. □

Problem 4. Continue to assume that b is strictly positive in $\bar{\Omega}$. Given $f \in L^2(\Omega)$, show that there exists a unique $u \in H^1(\Omega)$ such that

$$a(u, v) = \int_{\Omega} buv + \sum_{i,j=1}^n \int_{\Omega} a_{ij} \partial_i u \partial_j v = \int_{\Omega} f v$$

for all $v \in H^1(\Omega)$.

Proof. By the previous problem, $H^1(\Omega)$ is a Hilbert space when equipped with the inner product $(\cdot, \cdot) = a(\cdot, \cdot)$. For $f \in L^2(\Omega)$, the map

$$H^1(\Omega) \rightarrow \mathbb{R}, \quad v \mapsto \int_{\Omega} f v$$

is a bounded linear functional. The Riesz representation theorem thereby guarantees the existence of a *unique* function $u \in H^1(\Omega)$ such that

$$a(u, v) = (u, v) = \int_{\Omega} f v, \quad \forall v \in H^1(\Omega).$$

This completes the proof. $\square$

Problem 5. In the context of the previous problem, let $u \in C^2(\bar{\Omega}) \cap H^1(\Omega)$. Which classical problem does u solve?

Solution. By assumption, we have

$$\int_{\Omega} b u v + \sum_{i,j=1}^n \int_{\Omega} a_{ij} \partial_i u \partial_j v = \int_{\Omega} f v, \quad \forall v \in H^1(\Omega). \quad (20.2)$$

In particular, this holds for all test functions $v \in C_c^\infty(\Omega) \subseteq H^1(\Omega)$. For any function $v \in C_c^\infty(\Omega)$, we can integrate by parts to obtain

$$\int_{\Omega} b u v - \sum_{i,j=1}^n \int_{\Omega} v \partial_j (a_{ij} \partial_i u) = \int_{\Omega} f v.$$

Or rather,

$$\int_{\Omega} \left(b u - \sum_{i,j=1}^n \partial_j (a_{ij} \partial_i u) \right) v = \int_{\Omega} f v$$

for all $v \in C_c^\infty(\Omega)$. Thus, u will solve the problem

$$b u - \sum_{i,j=1}^n \partial_j (a_{ij} \partial_i u) \equiv f$$

in Ω . Next, we ask ourselves which boundary condition is imposed by the requirement that $u \in H^1(\Omega)$. Integrating by parts in (20.2) with $v \in C^1(\bar{\Omega}) \subseteq H^1(\Omega)$ yields

$$\int_{\Omega} b u v + \sum_{i,j=1}^n \left[\int_{\partial\Omega} a_{ij} v \partial_i u \nu^j - \int_{\Omega} \partial_j (a_{ij} \partial_i u) v \right] = \int_{\Omega} f v$$

whence it follows that

$$\sum_{i,j=1}^n \int_{\partial\Omega} a_{ij} v \partial_i u \nu^j = 0$$

for all $v \in C^1(\bar{\Omega})$. It therefore follows that $\sum_{i,j=1}^n a_{ij} v \partial_i u \nu^j = 0$ on $\partial\Omega$. $\square$

Problem 6. Let B be the open unit ball in $\mathbb{R}^n$ for $n \geq 3$ and let

$$q := 2^* = \frac{2n}{n-2}.$$

Show that the canonical embedding $H^1(B) \hookrightarrow L^q(B)$ is *not* compact.

Proof. Let φ be a non-trivial smooth function in $\mathbb{R}^n$ having compact support inside B . For a given $k \in \mathbb{N}$, we define a new function φ_k according to the rule:

$$\varphi_k : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \varphi_k(x) := k^{\frac{n-2}{2}} \varphi(kx).$$

Since $\text{supp}(\varphi) \subset B$, it is not hard to see that each φ_k is smooth with compact support in B . Thus, $\{\varphi_k\}_k$ is a sequence in $H^1(B)$. Letting $1 \leq p < \infty$ be given, we compute

$$\begin{aligned} \|\varphi_k\|_{L^p(B)}^p &= \int_{\mathbb{R}^n} |\varphi_k|^p = \int_{\mathbb{R}^n} k^{\frac{p(n-2)}{2}} |\varphi(kx)|^p dx \\ &= k^{\frac{p(n-2)}{2}} \int_{\mathbb{R}^n} |\varphi(kx)|^p dx \\ &= k^{\frac{p(n-2)}{2}} \cdot k^{-n} \int_{\mathbb{R}^n} |\varphi(x)|^p dx \\ &= k^{\frac{p(n-2)}{2}} \cdot k^{-n} \|\varphi\|_{L^p(B)}^p. \end{aligned}$$

Similarly,

$$\begin{aligned} \|\nabla \varphi_k\|_{L^p(B)}^p &= \int_{\mathbb{R}^n} |\nabla \varphi_k|^p = k^{\frac{p(n-2)}{2}} \int_{\mathbb{R}^n} |\nabla_x \varphi(kx)|^p dx \\ &= k^{\frac{p(n-2)}{2}} \int_{\mathbb{R}^n} k^p |\nabla \varphi(kx)|^p dx \\ &= k^{\frac{p(n-2)}{2} + p} k^{-n} \|\nabla \varphi\|_{L^p(B)}^p \\ &= k^{\frac{pn}{2}} k^{-n} \|\nabla \varphi\|_{L^p(B)}^p. \end{aligned}$$

Taking $p = 2$, we see that

$$\|\varphi_k\|_{L^2(B)}^2 = k^{n-2} \cdot k^{-n} \|\varphi\|_{L^2(B)}^2 = \frac{1}{k^2} \|\varphi\|_{L^2(B)}^2$$

which is bounded uniformly in k . Similarly,

$$\|\nabla \varphi_k\|_{L^2(B)}^2 = \|\nabla \varphi\|_{L^2(B)}^2$$

which does not depend on k . Hence, $\{\varphi_k\}$ is a bounded sequence in $H^1(B)$. On the other hand, taking $p = q = 2^*$, the above tells us that

$$\|\varphi_k\|_{L^q(B)}^q = k^{\frac{2n}{n-2} \cdot \frac{(n-2)}{2}} \cdot k^{-n} \|\varphi\|_{L^q(B)}^q = \|\varphi\|_{L^q(B)}^q \neq 0.$$

By way of contradiction, suppose that $\{\varphi_k\}$ has a subsequence in $L^q(B)$ that converges to some $f \in L^q(B)$. Let us keep the notation $\{\varphi_k\}$ for this subsequence. After passing to a subsequence, we might as well assume that φ_k converges pointwise almost everywhere to f on B . However, φ_k converges pointwise almost everywhere to 0. Hence, we must have $f = 0$ almost everywhere on B , which directly contradicts the fact that

$$\|f\|_{L^q(B)}^q = \lim_{k \rightarrow \infty} \|\varphi_k\|_{L^q(B)}^q = \|\varphi\|_{L^q(B)}^q \neq 0.$$

□

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